



Monitor and maintain a business computer system

D1.HGE.CL7.05 D1.HML.CL10.04
D1.HRM.CL9.13 D2.TRM.CL9.15

Trainee Manual



for foods, tourism
& hospitality

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Trainee Manual



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Introduction to trainee manual

To the Trainee

Congratulations on joining this course. This Trainee Manual is one part of a 'toolbox' which is a resource provided to trainees, trainers and assessors to help you become competent in various areas of your work.

The 'toolbox' consists of three elements:

- A Trainee Manual for you to read and study at home or in class
- A Trainer Guide with Power Point slides to help your Trainer explain the content of the training material and provide class activities to help with practice
- An Assessment Manual which provides your Assessor with oral and written questions and other assessment tasks to establish whether or not you have achieved competency.

The first thing you may notice is that this training program and the information you find in the Trainee Manual seems different to the textbooks you have used previously. This is because the method of instruction and examination is different. The method used is called Competency based training (CBT) and Competency based assessment (CBA). CBT and CBA is the training and assessment system chosen by ASEAN (Association of South-East Asian Nations) to train people to work in the tourism and hospitality industry throughout all the ASEAN member states.

What is the CBT and CBA system and why has it been adopted by ASEAN?

CBT is a way of training that concentrates on what a worker can do or is required to do at work. The aim is of the training is to enable trainees to perform tasks and duties at a standard expected by employers. CBT seeks to develop the skills, knowledge and attitudes (or recognise the ones the trainee already possesses) to achieve the required competency standard. ASEAN has adopted the CBT/CBA training system as it is able to produce the type of worker that industry is looking for and this therefore increases trainees' chances of obtaining employment.

CBA involves collecting evidence and making a judgement of the extent to which a worker can perform his/her duties at the required competency standard. Where a trainee can already demonstrate a degree of competency, either due to prior training or work experience, a process of 'Recognition of Prior Learning' (RPL) is available to trainees to recognise this. Please speak to your trainer about RPL if you think this applies to you.

What is a competency standard?

Competency standards are descriptions of the skills and knowledge required to perform a task or activity at the level of a required standard.

242 competency standards for the tourism and hospitality industries throughout the ASEAN region have been developed to cover all the knowledge, skills and attitudes required to work in the following occupational areas:

- Housekeeping
- Food Production
- Food and Beverage Service
- Front Office

- Travel Agencies
- Tour Operations.

All of these competency standards are available for you to look at. In fact you will find a summary of each one at the beginning of each Trainee Manual under the heading 'Unit Descriptor'. The unit descriptor describes the content of the unit you will be studying in the Trainee Manual and provides a table of contents which are divided up into 'Elements' and 'Performance Criteria'. An element is a description of one aspect of what has to be achieved in the workplace. The 'Performance Criteria' below each element details the level of performance that needs to be demonstrated to be declared competent.

There are other components of the competency standard:

- *Unit Title*: statement about what is to be done in the workplace
- *Unit Number*: unique number identifying the particular competency
- *Nominal hours*: number of classroom or practical hours usually needed to complete the competency. We call them 'nominal' hours because they can vary e.g. sometimes it will take an individual less time to complete a unit of competency because he/she has prior knowledge or work experience in that area.

The final heading you will see before you start reading the Trainee Manual is the 'Assessment Matrix'. Competency based assessment requires trainees to be assessed in at least 2 – 3 different ways, one of which must be practical. This section outlines three ways assessment can be carried out and includes work projects, written questions and oral questions. The matrix is designed to show you which performance criteria will be assessed and how they will be assessed. Your trainer and/or assessor may also use other assessment methods including 'Observation Checklist' and 'Third Party Statement'. An observation checklist is a way of recording how you perform at work and a third party statement is a statement by a supervisor or employer about the degree of competence they believe you have achieved. This can be based on observing your workplace performance, inspecting your work or gaining feedback from fellow workers.

Your trainer and/or assessor may use other methods to assess you such as:

- Journals
- Oral presentations
- Role plays
- Log books
- Group projects
- Practical demonstrations.

Remember your trainer is there to help you succeed and become competent. Please feel free to ask him or her for more explanation of what you have just read and of what is expected from you and best wishes for your future studies and future career in tourism and hospitality.

Unit descriptor

Monitor and maintain a business computer system

This unit deals with the skills and knowledge required to Monitor and maintain a business computer system in a range of settings within the travel industries workplace context.

Unit Code:

D1.HGE.CL7.05 D1.HML.CL10.04
D1.HRM.CL9.13 D2.TRM.CL9.15

Nominal Hours:

50

Element 1: Monitor effectiveness of computer equipment

Performance Criteria

- 1.1 Monitor and evaluate the effectiveness of computer equipment to ensure it meets enterprise needs through ongoing consultation with users
- 1.2 Monitor effectiveness through assessment of productivity and efficiency
- 1.3 Establish and implement procedures for identifying and resolving performance problems including use of specialised assistance
- 1.4 Identify computer training needs and provide training opportunities

Element 2: Purchase hardware and software

Performance Criteria

- 2.1 Identify broad computer equipment hardware and software requirements accurately and research possible new approaches using appropriate information sources
- 2.2 Assess the feasibility of acquiring or upgrading hardware and software based on available information
- 2.3 Evaluate options against the needs of the enterprise, including advantages and limitations
- 2.4 Investigate user-friendliness, system support and training issues
- 2.5 Seek external specialist assistance as required
- 2.6 Make decisions on computer acquisitions in consultation with appropriate colleagues
- 2.7 Plan introduction of new computer systems to take account of impacts on colleagues and customers
- 2.8 Ensure staff are adequately trained in using new systems
- 2.9 Monitor new systems for efficiency and take action accordingly

Element 3: Maintain computer equipment and systems

Performance Criteria

- 3.1 Establish and monitor systems for cleaning, minor maintenance and replacement of consumables in accordance with manufacturer's instructions
- 3.2 Report faults to the nominated person or supplier for rectification
- 3.3 Establish and monitor procedures for ensuring security of data, including regular back-ups and virus checks
- 3.4 Create document filing systems and procedures to reflect the size, nature and complexity of the workplace
- 3.5 Set up equipment to accept the software and functions being used

Assessment matrix

Showing mapping of Performance Criteria against Work Projects, Written Questions and Oral Questions

The Assessment Matrix indicates three of the most common assessment activities your Assessor may use to assess your understanding of the content of this manual and your performance - Work Projects, Written Questions and Oral Questions. It also indicates where you can find the subject content related to these assessment activities in the Trainee Manual (i.e. under which element or performance criteria). As explained in the Introduction, however, the assessors are free to choose which assessment activities are most suitable to best capture evidence of competency as they deem appropriate for individual students.

		Work Projects	Written Questions	Oral Questions
Element 1: Monitor effectiveness of computer equipment				
1.1	Monitor and evaluate the effectiveness of computer equipment to ensure it meets enterprise needs through ongoing consultation with users	1.1	1,2	1
1.2	Monitor effectiveness through assessment of productivity and efficiency	1.2	3,4	2
1.3	Establish and implement procedures for identifying and resolving performance problems including use of specialised assistance	1.3	5,6	3
1.4	Identify computer training needs and provide training opportunities	1.4	7	4
Element 2: Purchase hardware and software				
2.1	Identify broad computer equipment hardware and software requirements accurately and research possible new approaches using appropriate information sources	2.1	8,9,10,11,12	5
2.2	Assess the feasibility of acquiring or upgrading hardware and software based on available information	2.2	13	6
2.3	Evaluate options against the needs of the enterprise, including advantages and limitations	2.3	14	7

		Work Projects	Written Questions	Oral Questions
2.4	Investigate user-friendliness, system support and training issues	2.4	15,16	8
2.5	Seek external specialist assistance as required	2.5	17	9
2.6	Make decisions on computer acquisitions in consultation with appropriate colleagues	2.6	18,19	10
2.7	Plan introduction of new computer systems to take account of impacts on colleagues and customers	2.7	20,21	11
2.8	Ensure staff are adequately trained in using new systems	2.8	22,23	12
2.9	Monitor new systems for efficiency and take action accordingly	2.9	24	13
Element 3: Maintain computer equipment and systems				
3.1	Establish and monitor systems for cleaning, minor maintenance and replacement of consumables in accordance with manufacturer's instructions	3.1	25,26	14
3.2	Report faults to the nominated person or supplier for rectification	3.2	27	15
3.3	Establish and monitor procedures for ensuring security of data, including regular back-ups and virus checks	3.3	28	16
3.4	Create document filing systems and procedures to reflect the size, nature and complexity of the workplace	3.4	29	17
3.5	Set up equipment to accept the software and functions being used	3.5	30	18

Glossary

Term	Explanation
Client-Server Setup	A network setup where multiple clients (computers) connect to a single, central server.
Cloud storage services	Software and services that run on the Internet instead of your computer
Computer virus	Software programs that have been designed to spread from one computer to another and to negatively interfere with the operations of a computer.
Consumables	The items or materials that the assets consume during its normal operation.
CPU	Central processing unit - the main brain of your computer.
CRS	Computerised reservation systems
Desktops	Computers consisting of a base unit, monitor, keyboard and mouse.
DSL	Digital Subscriber Line. It is a technology for bringing high-bandwidth information to homes and small businesses over ordinary copper telephone lines
EFTPOS	Electronic funds transfer at point of sale. It is an electronic payment system involving electronic funds transfers based on the use of payment cards at payment terminals located at points of sale
Ethernet	A system for connecting a number of computer systems to form a local area network
Feasibility	An assessment of the practicality of a proposed plan or method
GDS	Global Distribution System
GHz	Gigahertz
Graphics card	Responsible for displaying images
Hard drive	A high-capacity, self-contained storage device containing a read-write mechanism plus one or more hard disks, inside a sealed unit
Hardware	The machines, wiring, and other physical components of a computer or other electronic system

Term	Explanation
Internet service provider	Organisation that provides services for accessing, using, or participating in the Internet
ISP	Internet service provider
Integration	The end result of a process that aims to stitch together different, often disparate, subsystems
IRR	Internal Rate of Return
IT	Information technology
LAN	Local area network. Supplies networking capability to a group of computers in close proximity to each other.
Laptops	Also called notebooks, laptops are portable computers that integrate the display, keyboard, a pointing device or trackball, processor, memory and hard drive all in a battery-operated package
LCD	Liquid crystal display
Leasing	The right to possess and use property for a fixed period of time
LED	Light-emitting diodes
Mb	Megabyte is a multiple of the unit byte for digital information
MFD	Multifunction devices
Modem	A device or program that enables a computer to transmit data over, for example, telephone or cable lines
Netbooks	These are ultra-portable computers that are even smaller than traditional laptops.
Network	A set of computers connected together for the purpose of sharing resources
Network card	A piece of hardware that allows computers to communicate over a computer network
NPV	Net Present Value
OSH	Occupational Safety and Health
Operating system	Software that runs your computer

Term	Explanation
Peer-To-Peer Setup	A network setup where every computer acts as both the client and the server
PC	Personal Computer
PM	Preventative maintenance
Preventative maintenance	Maintenance that is regularly performed on a piece of equipment to lessen the likelihood of it failing
Procurement	The process of sourcing and purchasing computer system assets
Resolution	The number of pixels (individual points of colour) contained on a display monitor, expressed in terms of the number of pixels on the horizontal axis and the number on the vertical axis
ROI	Return on investment
Router	A hardware device that allows a tourism establishment to connect several computers and other devices to a single Internet connection
Server	Computer that provides data to other computers
Software	General term for the various kinds of programs used to operate computers and related devices
Spam	Irrelevant or inappropriate messages sent on the Internet to a large number of recipients
Thin client	A very basic computer. Essentially just a screen, keyboard, mouse and network connection, it accesses centralised processing power across a network
Thumb drive	Another term for USB flash drive.
WAN	Wide area networks - which reach across cities and other countries
Warranty	A promise to repair or replace it if necessary within a specified period of time
WEP	Wireless encryption protocol

Element 1:

Monitor effectiveness of computer equipment

1.1 Monitor and evaluate the effectiveness of computer equipment to ensure it meets enterprise needs through ongoing consultation with users

Introduction

Whilst a key aspect of any travel and tourism business will still be attentive service provided by professional staff, one indispensable element of any successful organisation is through the establishment and use of key technologies.

A business computer system is one of the fundamental tools that tourism personnel use to provide services to customers.

With technology becoming a key component when researching, planning and booking tourism experiences, it is vital that an organisation has an effective computer system that will enable them to meet the demands of customers, the organisation, the industry and its varied suppliers.



In the tourism personalised service is provided through the unique blend of technology and physical service.

Not only must an organisation carefully select and use a suitable computer system, but that its staff are aware of how to use and manage each and every component to ensure efficient and effective service.

This manual will explain in more detail:

- The assets of a business computer system
- The benefits of having a business computer system
- Identifying the need for a business computer system
- Selecting and purchasing component of a business computer system
- Setting up a business computer system
- Establishing files, systems and procedures to best use a business computer system
- Training staff in skills required to operate and manage a business computer system
- Steps to recognise and resolve faults
- Establish maintenance programs.



Key assets of a business computer system

In Section 2.1 of this manual we will explore the different assets of a business computer system; however it is important to understand the different aspects of a computer system.

Key assets of a business computer system may include:

- Internet service provider
- Modem
- Router
- Network card
- Servers
- Wired or Ethernet networks
- Wireless networks
- Local area network (LAN)
- Network security
- Personal Computers (PC's)
- Monitors
- Keyboards
- Mouse
- Scanners
- Printers
- Multifunction devices (MFDs)
- Back-up systems.



Key expectations of a business computer system

Given that the purchase of a business computer system for any tourism business is a substantial investment, there are a number of expectations that must be met. It must be:

- Capable of performing the tasks you need to run a tourism organisation
- Affordable
- Well built
- Reliable
- Simple to use and maintain
- Easy to upgrade.





Activity 1 – Identify your expectations of a business computer system



In your own words, if you were a manager of a tourism organisation, outline the expectations you would have for a business computer system.

Understand business strategies and requirements

The use and purchase of computer assets are important activities in any business.

Before an organisation purchases an expensive computer system it is important to understand:

- What are the strategies of the organisation going forward
- What are the current and future business computer requirements.

There is a need to make sure the organisation has the technological assets it needs to achieve its identified business objectives, while at the same time, ensuring that money is not wasted on assets that are not necessary or inappropriate to the task.

Therefore, purchasing decisions about a computer system must be made carefully and cost-effectively. As technology changes and customer preferences change, there is a chance that some technologies may become obsolete at an alarming rate, whilst others will be essential in conducting day to day business in a competitive environment.

In getting a better understanding of what computer system requirements may be needed to support current and future business operations, a number of questions may need to be identified, to help guide the decision making process.



Develop business computer system research questions

These questions include, but not limited to:

- What are the key activities we currently perform?
- What business computer system assets are needed to perform what we currently perform?
- What business computer system assets are currently missing, that are essential?
- How is our current business computer system performing?
- What is the strategic plan for the organisation?
- What products and services will be provided to customers?
- What business computer system assets will be needed to provide these products and services?
- What new business computer system assets do we need to purchase?
- What existing business computer system assets can we retain?
- What existing business computer system assets can we update or upgrade?
- What physical office structure will be required to meet future strategic plans?
- How many staff will be required?
- What workspace area will they need?
- What is the current budget for the purchase of a business computer system?
- What will be the benefits of purchasing or upgrading our existing business computer system assets?
- How long will it take to get our money back?
- Will we lose money on this investment?
- How will we pay for the asset? Will it be financed? If so, how?
- Are there any non-financial reasons for wanting the business computer system assets?
- What assets are up for disposal, renewal or improvement?
- Is there a new business computer system asset that could boost or likely to increase revenue due to customer demand?





Activity 2 – Identify your own tourism operation

For the purpose of this manual, you are to identify a possible tourism operation that you would like to establish, FOR YOUR LOCAL AREA.

This can be an existing operation or a new fictional operation.

It is important to note, that the tourism operation you choose in this Activity, will be used for all activities in this manual.

You are required to provide a summary of your tourism organisation by identify the following:

- Name of tourism organisation
- Location of your tourism organisation
- Key products, services and activities provided
- Number of staff
- Summary of your computer system asset requirements.



Name of tourism organisation	
Location of your tourism organisation	
Key products, services and activities provided	
Number of staff	
Summary of your computer system asset requirements	

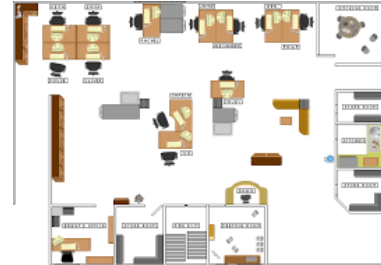


Activity 3 – Draw an outline of your office

For your tourism organisation identified in Activity 2, please draw an outline of your proposed office identifying the location of possible:

- Tables & Chairs
- Business computer system assets.

This outline will start to give you a visual understanding of what you will likely need for your business computer system.



Involving stakeholders in determining business computer system asset needs

While managers have the responsibility for making purchasing decisions regarding business computer systems, their actions need to reflect and address identified workplace needs.

This need may be immediate and urgent in nature or focussed on some future need that the business has.

The preferred way of establishing such need is to consult with the relevant stakeholders.

For managers it is important to identify:

- Who the relevant stakeholders may be
- Why their input is important
- How such contributions may be obtained.



Types of stakeholders

The composition of relevant stakeholders will vary depending on purpose, type and assets of a business computer system that is being considered for acquisition.

Potential stakeholders may include:

- Owners of the business – who can share their vision and expectations as well as their intention for the direction of the business
- Managers – who will be able to contribute information relating to the strategic direction of the business, financial limitations and other factors
- Supervisors – who will be able to contribute their thoughts on what is needed, how it will impact on operations and other practical matters
- Staff, teams and workgroups – who can provide input on the types of equipment that is needed for them to perform their tasks
- Customers – who should be asked about their thoughts, needs, wants and preferences
- Suppliers – who should be asked to contribute quotations, specifications, brochures, technical information, details of after-sales service (as appropriate) and names of other businesses who have already purchased the equipment and are currently using it
- IT consultants - they will be able to provide specialised advice on all elements of a business computer system and provide suggestions tailored to the technical needs of the organisation.



Why is their input important?

In most situations, it is best to obtain input from a wide variety of stakeholders so that a diversity of opinions and suggestions can be obtained.

It is possible that each stakeholder category will bring a unique and legitimate point-of-view to the deliberations. Without an appreciation of all pertinent factors, the possibility always exists that a flawed decision may be made.

All managers must guard against the belief that they, on their own, know best what is required. In most cases, consultation will indicate one or more factors that may only have been considered because of the consultative process undertaken.

Involving stakeholders in the process of identifying significant resource requirements can be time-consuming but it is beneficial in that it can help determine:

- The structure of the business computer system to meet the desired technical needs
- Identification of all pieces of hardware and software assets required
- Quantities to be obtained
- Timelines that apply to acquisition
- Brands, models and types to be considered
- Restrictions that apply – such as preferred supplier arrangements, budget, compatibility, space issues, compliance requirements
- The needs of those who will be using the equipment
- Customer needs, wants and preferences
- Introduction and installation requirements
- Training required
- Financing arrangements.





Activity 4 – Identify stakeholders

You are asked to identify the different stakeholders you would consult with when planning an appropriate business computer system.

For each of the identified stakeholders, list the ways in which their contribution can be useful or influential in the identification of computer system asset requirements.



STAKEHOLDER	TYPE OF INPUT REQUIRED	METHOD TO OBTAIN INPUT

1.2 Monitor effectiveness through assessment of productivity and efficiency

Introduction

As manager it is your overall responsibility to see that systems are set up to ensure that the condition and performance of business computer systems and regularly monitored, reported and discussed within the enterprise.

Many assets purchased by a business, including computer system assets will need to have their performance monitored.

Monitoring performance allows more informed decisions to be made about the effectiveness of the item in relation to the purpose for which it was acquired, and to guide future purchasing decisions when replacement or supplementary items need to be obtained.

It is important to understand the possible processes that can be used to determine resource use and performance.

Before we look at the need to purchase or upgrade a computer system, it is important to monitor and evaluate the effectiveness of the current computer system currently being used by a tourism organisation.



Computer system asset management

Computer system asset management, is the way in which we look after the computer assets in our organisation, either day-to-day (maintenance and operations) or in the medium to long term (strategic and forward planning).

Computer system asset management is the process of guiding the acquisition, use and disposal of business computer system assets to make the most of their service delivery potential and manage the related risks and costs over their entire life'.

Business computer system asset management is directly associated with the following:

- The current or desired levels of service of our customers
- The associated costs in providing these services
- The practices and systems that assist in achieving this in the most effective and efficient ways.

Managers now find themselves involved in all or some of the following activities:

- Planning for assets
- Creating or acquiring new assets
- Accounting for assets and determining the actual cost of the services they provide
- Operating and maintaining assets
- Monitoring performance and condition of assets
- Assessing refurbishment, renewal and replacement options



- Disposing of assets that are no longer required
- Auditing the way in which assets are managed and the asset management practices and procedures
- Identifying and working to minimise life cycle costs of new assets.

Computer system asset monitoring system

An asset monitoring system collects and reports information on the condition and performance of computer assets.

The performance of the asset is evaluated against the level of service provided to the customer. This can generally be measured in terms of reliability, availability, capacity and meeting customer demands and needs.

Typical questions that may be answered by effectively monitoring the assets include:

Condition assessment

- Where is the asset in its life cycle?
- When was the asset constructed, last replaced or refurbished?
- What is the asset's effective or expected life?
- What is the residual life of the asset before:
 - Replacement is necessary?
 - Refurbishment is required?
- Has the asset been inspected and by whom, using what process?
- Can the asset's deterioration be measured and predicted?
- Can the asset's failure be predicated? If so, how?
- Can planned maintenance prevent the asset's failure or extend the time before failure?
- Can the asset be refurbished?
- What will it cost to refurbish the asset?
- What level of service will the asset deliver once it is refurbished?
- Is the asset technically or commercially obsolete?



Performance monitoring

- Is the asset performing adequately for:
 - Reliability?
 - Demands on its use or capacity?
 - User needs?
- What is the asset's current utilisation rate compared to its total capacity? Is it below, at or above its capacity?
- Have standards of service been set for each type of asset?
- Is the performance of the asset being reported to the people who made the acquisition decision?



- Does the asset meet:
 - Occupational health and safety regulations?
 - Public safety requirements?
 - Environmental requirements?
- What is the risk of the asset's failure?

Systems to monitor condition and performance of business computer system

Systems to monitor condition and performance of business computer systems may include:

- Integration of reporting into day to day operating procedures
- Regular management reports
- Internal and external inspections and audits
- Regular staff feedback
- Analysis of maintenance costs over a period of time.





Activity 5 - Business computer system asset performance

You are to outline systems you would use to condition and performance of a computer system tourism organisation.



monitor
for your

Types of systems	Strategies / examples for each system
Integration of reporting into day to day operating procedures	
Regular management reports	
Internal/external inspections and audits	
Regular staff feedback	
Analysis of maintenance costs over a period of time.	

Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

1.1 Please complete the following activities relating to this Performance Criteria:

- Activities 1,2,3,4

1.2 Please complete the following activities relating to this Performance Criteria:

- Activity 5

1.3 Please complete the following activities relating to this Performance Criteria:

- Activities 26,27

1.4 Please complete the following activities relating to this Performance Criteria:

- Activity 21
-

Summary

Monitor effectiveness of computer equipment

Monitor and evaluate the effectiveness of computer equipment to ensure it meets enterprise needs through ongoing consultation with users

- Key assets of a business computer system
- Key expectations of a business computer system
- Understand business strategies and requirements
- Involving stakeholders in determining business computer system asset needs

Monitor effectiveness through assessment of productivity and efficiency

- Computer system asset management
- Computer system asset monitoring system
- Systems to monitor condition and performance of business computer system

Element 2: Purchase hardware and software

2.1 Identify broad computer equipment hardware and software requirements accurately and research possible new approaches using appropriate information sources

Introduction

One of the most important tasks in deciding on the purchase of a business computer system is to understand the different types of assets that make up a system.

Managers will be responsible for selecting and applying the appropriate technology and software applications to produce workplace documents and conduct a number of technology based activities.

In today's world, there are endless types of components, made by various companies, with different features, provided at different levels of quality and performance.

It is essential that the computer system assets purchased are able to meet the identified needs of the organisation, as identified in Section 1.1.

It is vital that a tourism organisation understands:

- The different assets available
- The purposes of each asset
- Key features of each asset
- Different suppliers providing the asset
- The cost of the asset
- Additional services, guarantees, warranties etc. associated with the asset.



Whilst each tourism operation will have its own preferred business computer system assets, common assets will be identified in this section.

Sources of computer system information

As part of the process of identifying and selecting appropriate business computer assets, it is wise to conduct research using a variety of specialist technology experts and sources.

When setting up a business computer system, it is a good idea to involve the services of professionals as they will be able to design and set up the best system to meet the needs of your organisation.

Excellent information sources relating to new technology and computer assets may include:

- Computer magazines and journals
- Trade fairs
- Computer software designers
- Computer hardware manufacturers
- Retail outlets
- Internet
- Industry associations
- Seminars, workshops and training sessions
- Internal/external clients.



Setting up a computer network / system

Setting up a computer network is a good way to get more out of your system and its assets.

This is especially true if for a smaller tourism establishment.

Networking allows personnel to manage all of the operations of a small business including communication, data transfer, and storage.

By setting up a network, a computer will be used to its full potential and all your business needs will be met.

It enables access to the internet with any computer on the network, as well as share files and devices across it. An effective computer network design can make a big difference for a tourism business.

Internet connectivity

Key requirements to establish internet connectivity include:

Internet service provider (ISP)

Common types of internet service:

- Dial-up - Dial-up is generally the slowest type of Internet connection
- DSL - DSL service uses a broadband connection, which makes it much faster than dial-up. DSL connects to the Internet via phone line. Unlike dial-up users can use the Internet and your phone line simultaneously
- 3G and 4G - 3G and 4G service is most commonly used with mobile phones and tablet computers, and it connects wirelessly through your ISP's network.

Considerations when selecting an ISP include:

- Speed
- Price
- Ease of installation
- Service record
- Technical support
- Contract terms.



Modem/A DSL modem

The primary piece of hardware needed to connect to the internet is a modem.

The type of Internet access selected will determine the type of modem an organisation will need including:

- Dial-up access uses a telephone modem
- DSL service uses a DSL modem.

Router/A wireless router

A router is a hardware device that allows a tourism establishment to connect several computers and other devices to a single Internet connection, which is known as a network. Many routers are wireless, allowing an organisation to easily create a wireless network.

You don't necessarily need to buy a router to connect to the Internet.

It's possible to connect a computer directly to your modem using an Ethernet cable. Also, many modems now include a built-in router, so you have the option of creating a network without having to buy more hardware.

Most routers also act as a hardware firewall, which helps prevent others from gaining access to your computer through the Internet.



Network card

A network card is a piece of hardware that allows computers to communicate over a computer network. Most new computers have a network card built into the motherboard, so it probably isn't something you'll need to purchase. The network card will have an Ethernet port, a wireless connection, or both.

If you have a laptop with a wireless connection, you can access the Internet at any place that offers a Wi-Fi connection. Many restaurants, coffee shops, bookstores, hotels, and other businesses offer free Wi-Fi. In addition, many cities provide free Wi-Fi in public areas such as parks and downtown areas.

Servers

A computer that has been optimised to provide services to other computers over a network

Servers usually have powerful processors, lots of memory and large hard drives.

Setting up your Internet connection

Once you have chosen an ISP and purchased the appropriate modem, you can use the instructions provided by your ISP, or included with the modem, to set up your Internet connection



Wired and Wireless Networks

The first decision you will need to make about your new computer system network is whether you would like it to be wired or completely wireless.

Wired networks are more suited towards desktop PC setups whilst wireless networks are more suited to laptop use.

Companies are increasingly enabling their local networks for Wi-Fi wireless access, although larger businesses also tend to wire their office buildings with high-speed Ethernet cabling for greater network capacity and performance.

Where tourism organisations use a combination of desktops and laptop computers, a combination of the two can be used. For example:

- Desktop - connected to the Internet with an Ethernet cable
- Laptop computer - connected via a wireless router.

These two methods will have their advantages and disadvantages, but either one will be suitable for a tourism organisation.



Wired or Ethernet networks

- Extremely reliable
- Economical
- Secure
- Easy to install
- Use of wires to connect components.

Wireless networks

- Greater flexibility
- Broadband access from remote locations
- Very easy to install
- Eliminate the need for wires or cords.



Local area network (LAN)

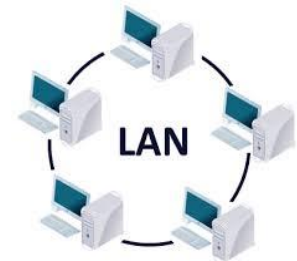
A local area network (LAN) supplies networking capability to a group of computers in close proximity to each other.

A LAN is useful for sharing resources like files, printers, games or other applications

Most local area networks use Wi-Fi or Ethernet for connectivity between devices.

Each of these local networks may:

- Stand-alone disconnected from any other network
- May connect to other LANs
- Connected to a WAN (wide area networks) - which reach across cities and other countries.



Small office networks normally function with either one or two local area networks (LANs), each controlled by its own network router.

Network Setup

The next step in how to set up a small business computer network is deciding whether to make it a peer-to-peer setup or a client-server one. Both networks connect computers so that resources can be shared between them. The fundamental differences are in the setup configuration.

Peer-To-Peer Setup

In a peer-to-peer setup, every computer acts as both the client and the server. Each computer communicates directly with the other computers in the network and resources can be added or removed. A peer-to-peer setup is much more common in the home.

Client-Server Setup

In a client-server setup, multiple clients (computers) connect to a single, central server. Public data and applications are only installed on the server and the clients connect to the server to use the resources. This type of setup is more typical in larger offices or businesses.

Setting up a secure network

Once you have decided upon the type of network, it is important that it is protected.

You can secure a network a number of ways, including through:

- The router - through the use of security settings
- The operating system.

You can do this through technologies such as wireless encryption protocol (WEP) for a wireless network, which uses passwords to encrypt the network traffic.

You may also want to use Windows and network logins and passwords to limit access to the network to authorised users. You may also choose to use hardware security like fingerprint recognition, security and password keys, and full disk encryption to further protect the network. When it comes to using the Internet, businesses should also check their browser security and privacy settings to ensure that the network is protected from viruses, spam and hacking attempts.

Personal Computers (PC's)

Personal computers are an essential part of any business computer system.

The personal computer (PC) defines a computer designed for general use by a single person.

The two main operating systems used for PC's include:

- Windows
- Mac.

Types of personal computers

Following are the main types of personal computers.

Desktops

Desktop computers consist of a base unit, monitor, keyboard and mouse.

They are designed to be set up and used in one permanent location.

Most desktops offer more power, storage and versatility for less cost than their portable brethren.



Laptops

Also called notebooks, laptops are portable computers that integrate the display, keyboard, a pointing device or trackball, processor, memory and hard drive all in a battery-operated package.

Netbooks

These are ultra-portable computers that are even smaller than traditional laptops. The extreme cost-effectiveness of netbooks means they're cheaper than almost any brand-new laptop. However, netbooks' internal components are less powerful than those in regular laptops.

Tablet computers

These are touch screen devices that are very portable and versatile.

Thin clients

If you need to purchase a number of computers in your business at once, you might consider a thin client system instead of buying individual business computers.

A thin client is a very basic computer. Essentially just a screen, keyboard, mouse and network connection, it accesses centralised processing power across a network.

Using a thin client feels just like using a normal computer. The difference is that all the work is being done by a central server.



Key features of a personal computer

There are a number of key features of a PC.

These will be explained in more detail below.

Processor

Also called the central processing unit (CPU), this is the computer chip that is the main brain of your computer.

The processor determines the speed of the computer.

CPUs are made by either Intel (which makes Celeron, Centrino and Pentium) or AMD (maker of Sempron, Athlon and Turion). Celeron and Sempron are usually found in lower-spec budget PCs.

Two main factors determine a CPU's performance: the number of cores and the speed at which it runs. The speed is measured in gigahertz (GHz).

A dual or quad core chip is effectively two or four CPUs in one. This makes your computer more effective when it runs several programs at once.



Make sure your new business computers have at least a dual core processor running at a minimum of 2.5GHz.

Graphics card

This is also known as a graphics chip.

The graphics card is most important if you want to play games.

This is responsible for displaying images. Integrated graphics means the chip shares the computer's Ram, which makes it slower.

Graphics cards usually have their own dedicated Ram – the amount you need depends on what you plan to do but try to ensure you have at least 256MB if you're buying a desktop or 128MB for a laptop.

The graphics card is most important if you want to play games. To get the most out of the most up-to-date games being released, make sure your graphics card is DX 11 compatible.

Hard disk

This is also called a hard drive.

Most PCs come with 300GB or more

This is where your computer stores all your software and files. The size is measured in gigabytes (GB). Try to ensure any laptop you buy has at least 250GB.



Most desktop PCs come with 500GB or more, which should be plenty – but bear in mind what you'll be using your computer for.

Operating system

This is the software that runs your computer. Macs come with OS X, and PCs usually come with Windows.

Macs come with the Mac OS operating system.

RAM

Random Access Memory or Memory - the more Ram a computer has, the better it will run. The minimum recommended is 2GB of RAM.

Monitors

Types of computer monitors

All staff will use computer monitors daily at work. And while these come in a variety of shapes, designs, and colours, they can also be broadly categorised into two popular types.

LCD (liquid crystal display) monitors

The LCD monitor incorporates one of the most advanced technologies available today.

The advantages of LCD monitors include:

- Compact size
- Lightweight
- Don't consume much electricity
- Images don't get geometrically distorted.

LED (light-emitting diodes) monitors

LED monitors are the latest types of monitors on the market today. These are flat panel, or slightly curved displays

The advantages of LED monitors are:

- They produce images with higher contrast
- Are more durable than a LCD monitor
- Features a very thin design
- Don't produce much heat while running.



Considerations when purchasing monitors

These include:

- Size - can be one of the more important considerations when purchasing a computer monitor
- Resolution - is the maximum resolution of pixels capable of being shown horizontally x vertically.

As the monitor increases in size, the image display will decrease; below is a listing of a usable resolution limit by the size of the monitor. These sizes can be debated but are the standard and most viewed resolutions.

SIZE	RESOLUTION
15"	800 x 600 or 1,024 x 768
17"	1,280 x 1,024
19"	1,280 x 1024
21"	1,600 x 1,200

Keyboards

Keyboards are available in a wide array of options, with some suitable for use only with nominated or 'supported' software.

More and more keyboards are wireless.

Options available in keyboards can include:

- Ergonomic design for safer and more comfortable use including integrated palm rest or detachable palm rest
- Customisable hot keys
- Single touch pad
- Keyboard scrolling wheel
- USB ports.



Mouse

The mouse now also comes with a variety of features, including a wireless option.

The shape of the mouse has changed quite a deal over the years with current designs claimed as being more ergonomic and having better grip.

Further features include:

- Extra buttons – enabling customisation to commonly used keystrokes or Internet commands
- Wheel scrolling for accelerated page scrolling
- Left-hand and right-hand design.

Scanners

Scanners have been described as 'photocopiers for computers'.

They enable pictures and text from an existing hard copy source to be 'scanned' into the computer.

Photos, logos, 3-D objects, and graphics can be imported into a computer file for use or amendment.

It also enables hard copy text to be scanned into a file, thus saving the time consuming task of typing it in.

Some scanners will not only scan, but also copy, e-mail and post to the web.



Printers

Most establishments will probably use a black and white and colour laser printer.

Modern printers boast 'photo quality' copies. Six colours are common and some printers have easy print screen function, borderless copies, ink backup printing mode, duplexing facility, glossy copy and digital photo options (direct from camera).

Multifunction devices (MFDs)

MFDs are pieces of equipment where a number of items are combined into the one piece of equipment.

The common functions are performed using MFD's include:

- Print
- Copy
- Scan
- Fax.



Computer back-up systems

With the amount of data collected and stored by a tourism organisation, it is essential that not only is the data backed up to ensure it is not lost, but also to transfer data from PC's to back up systems to speed up applications and processes.

When choosing a backup system there are a number of considerations including:

- Amount of data you have
- Amount of computers
- Amount of hands-on involvement required
- Backup speed
- Sensitivity of your data
- Your level of computer expertise
- Physical size of the drive.

Types of back-up drives

Backup solutions include:

- Internal drives
- External drives
- Network drives
- Online services
- Thumb drives
- CD or DVD backups.





Activity 6 - Identify business computer system hardware assets

You are required to prepare a list of business computer system hardware assets that you would require for your tourism business.

Refer to your office plan in Activity 3 to help guide your identification process.

Please complete this document in a workable word document as it will be used in later activities.



TYPE OF HARDWARE ASSET	DESCRIPTION OF YOUR DESIRED REQUIREMENTS/BRANDS/SIZES	REQUIRED QUANTITY	CURRENT USABLE QUANTITY	REQUIRED QUANTITY TO ACQUIRE	POSSIBLE SUPPLIERS		
Internet service provider							
Modem							
Router							
Network card							

TYPE OF HARDWARE	DESCRIPTION OF YOUR DESIRED REQUIREMENTS/BRANDS/SIZES	REQUIRED QUANTITY	CURRENT USABLE QUANTITY	REQUIRED QUANTITY TO ACQUIRE	POSSIBLE SUPPLIERS		
Servers							
Wired or Ethernet networks							
Wireless networks							
Local area network (LAN)							
Network security							
Personal Computers (PC's)							

TYPE OF HARDWARE	DESCRIPTION OF YOUR DESIRED REQUIREMENTS/BRANDS/SIZES	REQUIRED QUANTITY	CURRENT USABLE QUANTITY	REQUIRED QUANTITY TO ACQUIRE	POSSIBLE SUPPLIERS		
Monitors							
Keyboards							
Mouse							
Scanners							
Printers							
Multifunction devices (MFDs)							

TYPE OF HARDWARE	DESCRIPTION OF YOUR DESIRED REQUIREMENTS/BRANDS/SIZES	REQUIRED QUANTITY	CURRENT USABLE QUANTITY	REQUIRED QUANTITY TO ACQUIRE	POSSIBLE SUPPLIERS		
Back-up systems							
Other							



Software packages

To date in this section we have looked at the 'hardware' elements of a business computer system.

Whilst this is an essential element of any computer system, it is important to understand that software is just as important as hardware.

In general, software is a general term for the various kinds of programs used to operate computers and related devices.

The best software for tourism organisations is software designed to help them to run their operations better, cut costs and replace paper processes.

The most popular software packages include accounts, office productivity, email and communications, but nowadays, most business activities can be improved through desktop or web-based applications.

Word processing

Word processing packages are the most commonly used type of computer software package.

In simplistic terms they allow text to be typed and printed on paper.

However, a word processing package is much more than just an automated typewriter.



The most prominent difference is that anything typed on a word processor can be saved. It can then be reloaded, modified if necessary, and printed again. As a result, pages or even entire documents do not have to be retyped because of typing errors or other small changes.

Spreadsheets

After word processing packages, spreadsheets are probably the most widely used software application.

Spreadsheets are designed primarily to manipulate numbers and the main advantage in using one comes from a feature known as automatic re-calculation.

When calculating something on paper, changing one figure at the beginning of the problem would mean that each figure dependent on it would have to be re-calculated.

If the same problem was set up on a computer spreadsheet, and a figure changed, all re-calculations would be made automatically.



Databases

Databases allow information to be stored, sorted and selectively retrieved.

A structure is created to store data by specifying names and data types for each category of information.

Once the data has been entered, the speed and accuracy of the computer is used to process it in various ways.

The questions can range from the simple to more complex ones involving multiple search criteria. The advantage in using the database stems from the speed and accuracy with which the computer can answer the question.

Communication

E-mail allows users at different locations to send text messages to each other. Communications may be between adjacent rooms over a local area network (LAN) or between users at remote locations over the Internet.

Presentation

These packages are designed to create electronic presentations consisting of a series of separate pages or slides.



Sharing

A computer program for free-form information gathering and multi-user collaboration; it gathers users' notes (handwritten or typed), drawings, screen clippings and audio commentaries.

Internet browsers

These are primarily intended to use the World Wide Web; they can also be used to access information provided by web servers in private networks or files in file systems.

Accounting software

Few businesses can do without finance and accounting software, which can often automate activities such as billing, invoices and payroll.

Accounting software provides a way of recording and measuring sales, costs and profits. It can also be used to automatically generate invoices and formal business correspondence.

Some accounting software allows you to carry out analysis on your figures, to see trends and patterns in your business' performance

Groupware and collaboration

This type of software has its merits and can facilitate teamwork, and increase business process efficiencies.

These tools can help to facilitate electronic communications, so staff can send messages, files, data, or documents more easily, and even send and share voice files.

The sorts of functions that the software can bring are online conferencing, electronic faxing, interactive voice mail, wikis, web publishing and revision control of documents

Integrated packages

Today, most of these software packages are integrated providing all of the above applications.

One of the best known integrated packages is 'Microsoft Office' which includes the following packages:

They include the following packages:

- Word processing – Microsoft Word
- Spreadsheet – Microsoft Excel
- Data – Microsoft Access
- Communications – Microsoft Outlook.
- Presentation - PowerPoint
- Sharing - OneNote
- Internet browser – Microsoft Explorer.



The advantage is that it is relatively easy to transfer data, information or graphics from one application to another.

Tourism computer systems

In this section, we have looked at the different types of business computer system assets. It is important to understand the importance of tourism related computer systems.

Computerised reservation systems have become an essential part of any tourism organisation. It is a necessary tool to store and access a wide range of information that can be easily accessed by a range of people in any location.

The use of computerised reservation systems describes a key sales and operational function for a diverse range of domestic and international tourism products and services and applies to the full range of industry sectors.

Tourism organisations use a diverse range of computerised systems to manage reservations, operations and sales administration functions, so the system will vary depending upon the organisation and industry sector.

Commonly used industry-wide systems used by travel organisations include:

- Global distribution system (GDS)
- Computerised or centralised reservations system (CRS).

It covers systems used by tour operators when receiving and processing reservations for the supply of their product or service.

It also covers other industry-wide systems used by inbound tour operators, outbound tour wholesalers and meetings and events management organisations.

It can equally apply to any organisation-based computerised reservations or operations system.



The range and complexity of products and services and the particular reservations or operations system will vary according to the type of tourism operation.

Global Distribution System (GDS)

The Global Distribution System (GDS) is an e-commerce tool introduced by airlines to facilitate the booking of flights.

By definition a GDS is a computerised reservation system (CRS) in that it enables, for example, travel agencies to place bookings with travel suppliers and their booking systems via a worldwide distribution network.

There are four major Global Distribution Systems:

- Amadeus
- Galileo
- Sabre
- Worldspan – merged with Galileo in late 2006.



These systems enable travel agencies to:

- Search lowest fares for nominated destinations
- Automatically re-calculate fares for changed itineraries
- Store client data and records
- Access world-wide options almost instantaneously
- Have the capacity capable of booking:
 - One way and roundtrip airline seats
 - Hotel rooms
 - Rental cars
 - Tours
 - Cruises
 - Bus and rail tickets
 - Insurance
 - Limousines
 - Event and theatre tickets
 - Dining reservations.



Computerised reservation systems (CRS)

There are many computerised reservation systems (CRS) in use today and while they all provide similar functions, menus, sub-menus, options, reports, statistics and accounting facilities the individual programs are all unique.

This means that the specific operational aspects of each CRS will differ, including:

- The way you log on
- The terminology used
- The classifications provided for
- How you enter guest information
- Move between screens
- Create new files
- Manipulate information
- Modify and retrieve booking
- Print accounts and reports.

Benefits of a computerised reservation system

Benefits of using a CRS include:

- Speedier processing of requests and bookings, ticketing and quotations
- Central location of information, after data has been initially loaded onto the system, immediate access to comprehensive information on clients and allied reservation details are available through links via one screen; examples include:
 - Destination information, currency, weather, attractions, transport options, languages, time zone
 - costs of any product/service, including options, packages, deals, special offers and conditions
 - Airfares, including full flight details such as departure times, flight times, aircraft configuration, aircraft type/details
 - Airport taxes, application, amount/s and whether included or additional
 - Availability of products or services, including information such as minimum stay/quantities, seasonal price fluctuations, peak and low times
 - Vehicle details, including details about engine capacity, seating, options, rear or front wheel drive
 - Touring inclusions, describing the 'side' trips, cruises, transfers, meals, upgrades, visits and experiences included in the price
 - Product information, providing extensive detail about a wide range of products and services such as brand names, styles, colour, dimensions, weight, options
 - Payment requirements, explaining the deposit and full amount, payment options, due date and refund conditions
 - Health recommendations/pre-requisites, bookings such as action holidays, require a certain level of health/fitness of participants
 - Customs and immigrations, detailing the items that must be declared, the legal movement of items between countries and the requirements and limitations that apply to passports, visas and permits
- Integration with web-based/online booking systems
- Allowing multiple uses and multiple sites
- Preserving privacy and confidentiality through passwords, operator only designations and system administrator status
- Allowing pre-set limits/allocations/changes, such as dates, times, maximum room numbers, maximum bookings to be programmed into the system.



Barriers of a computerised reservation system

Barriers of a CRS include:

- Cost of initial establishment
- Training of staff
- System breakdowns and malfunctions
- Need for system back-ups, system maintenance and system updates
- Discrepancies occurring between properties operating a manual reservation system and the computerised reservations system
- Operational staff tend to focus on the screen/system as opposed to being customer-focused.



It is quite clear to see that there are many reasons why many tourism organisations will use a CRS system to help them manage their businesses.



Activity 7 - Identify business computer system software assets

You are required to prepare a list of business computer system software assets that you would require for your tourism business.

Refer to your office plan in Activity 3 to help guide you identification process.

Please complete this document in a workable word document as it will be used in later activities.



TYPE OF SOFTWARE ASSETS	DESCRIPTION OF YOUR DESIRED SOFTWARE REQUIREMENTS	REQUIRED QUANTITY	POSSIBLE SUPPLIERS		

2.2 Assess the feasibility of acquiring or upgrading hardware and software based on available information

2.3 Evaluate options against the needs of the enterprise, including advantages and limitations

Introduction

In the previous section we have explored all the possible hardware and software asset requirements for a business computer system.

Now that we have identified the types of computer assets and possible quantities required it is now time to undertake procurement.

As part of the procurement process a feasibility study will be conducted to identify, research, evaluate and select the best computer assets to meet your requirements.



Procurement

Procurement is the process of sourcing and purchasing computer system assets.

The task of procurement is not simply about identifying suitable products, services and suppliers but also requires expertise and experience to sourcing decisions that consider for example:

- Cost
- Availability
- Warranties and after-sales service
- Lead times
- Service centre locations
- Interruptions related to servicing
- Parts and replacement policies
- Product quality
- Reliability, suitability of use and durability
- Continuity of supply
- Terms of trade
- Rebates and discounts
- Staff training provided by suppliers
- Specials, deals, bonuses, prizes and other incentives to buy.



Procurement is also about the integrity and quality of the supplier and the ability to develop a long-term relationship with them.

Companies that focus on price at the expense of developing a quality supplier relationship often find that they have little loyalty paid back when issues arise such as product shortages or problems with computer spare parts or speed of after sales service.

The buyer–supplier relationship is also a valuable source of industry intelligence providing inside information on how other businesses are performing by virtue of the types of assets they are purchasing, and on the basis of personal observations made by sales representatives and service staff etc.



Definition of feasibility

Before we look at the tasks associated with conducting research to identify what is possible, it is important to understand the term 'feasibility'.

In summary, feasibility is based around objectively and rationally researching and understanding the strengths and weaknesses of various options to make an informed and calculated decision.

Feasibility activities

As part of making an informed decision regarding acquiring required business computer system assets, there are a number of activities to be undertaken.

Assess the feasibility may relate to:

- Identify the best method of acquiring required components
- Identifying advantages and disadvantages of each acquisition method
- Researching the different options that exist for the desired computer asset
- Assessing costs against available budget.



Methods to acquire computer system assets

Financial management in the organisation usually have a major influence over the decision to buy computer system assets. When a request is made for money to purchase or support these assets, management and the finance department will judge the worthiness of each asset on its own merits, rank it relative to other options, and provide authorisation for purchase.

Any proposal for a computer asset acquisition must detail the costs and benefits of that investment.

Do the benefits of the investment outweigh the costs? In the initial phase, the determination of the need, the classification of asset to be purchased needs to be identified.

Off-the-shelf products

- Selecting what the supplier produces as part of the product line
- Off-the-shelf generally has shorter lead time
- Greater availability of spare parts
- Greater experience on issues such as maintainability and reliability
- Warranty issues are clear as to the liability of the supplier.



Custom built

- Tailor-made directly to your requirements
- Longer lead times
- Possible limitations on availability of spare parts
- Limitations on reliability and cost issues
- Limitations on cost control and warranty issues
- Possibly lower resale value.

Used equipment

In some cases, the purchase of new equipment is not necessarily essential. There may be existing computer assets, either owned by the organisation or available second hand, which may be suitable.

Considerations include:

- Is it obsolete
- Expected remaining 'productive life'
- Cost savings vs. benefits
- Expected maintenance costs
- Availability of parts
- Existing service warranties
- If it needs reconditioning.





Activity 8 - Identify advantages and disadvantages of different classification of assets

For your tourism organisation you are to identify the advantages and disadvantages of acquiring the following classifications of computer assets:

- Off the shelf
- Custom built
- Second hand / refurbished.



ADVANTAGES	DISADVANTAGES
Off the shelf	
Custom built	
Second hand / refurbished	



Activity 9 - Identify computer assets to be acquired using each of these different classification of assets

Based on the list of hardware and software assets identified in Activities 6&7, identify examples of assets that you would obtain:

- Off the shelf
- Custom built
- Second hand / refurbished



ADVANTAGES	DISADVANTAGES
Off the shelf	
Custom built	
Second hand / refurbished	

Conduct research into required computer systems assets

The whole purpose of this process is to gather current, accurate and relevant data used for:

- Identifying the most suitable asset
- Estimating costs for the asset
- Understanding support services and warranties etc.

This information can be gathered from:

- Quotations from suppliers - gathered through the tender process (see below)
- Current maintenance contracts
- Published or advertised prices
- Previous contracts and costs.



The tender process

Collect tender information

Tender information collected by suppliers includes, but is not limited to:

- Clear precise details
- Product name and description
- What it is to be used for and the desired rating or production requirements
- Costs
- Storage procedure or installation standards
- Delivery required and ordering procedure
- Warranties.



Evaluation of tender information

Evaluating information required from suppliers may focus on:

- Quality standards
- Availability
- Delivery schedules
- Order quantities
- Product knowledge
- References
- Services
- Promotional materials.



Rating essential and negotiable requirements

- Availability of goods
- Delivery service meeting requirements of establishment
- Availability of suppliers
- Minimum/maximum order quantities
- Financial terms available - discounts and periods of credit
- Suppliers record
- Specialty knowledge
- References from other customers
- After sales service.



Identify criteria used to evaluate delivery response times by supplier

- Is the product readily available?
- Can you store enough products to survive delay?
- Trading hours of the supplier?
- How flexible is the supplier going to be?



Activity 10 - Identify preferred hardware and software asset required

Based on the computer system assets required for your tourism organisation, you are to conduct research and identify which is the most suitable type/description/brand for each hardware and software asset.





Activity 11 - Obtaining information from suppliers

Now that you have identified the most suitable type/description/brand for each hardware and software asset, you will now collect information and quotations from different suppliers.

You are required to:

- Collect a minimum of three quotations from possible suppliers
- Outline the key advantages and disadvantages of using this supplier



The following is a template that you may use for each and every required computer system asset.

ASSET: _____

ASSET TYPE/DESCRIPTION/BRAND _____

SUPPLIER 1: _____ COST: _____	
Advantages	Disadvantages
SUPPLIER 2: _____ COST: _____	
Advantages	Disadvantages
SUPPLIER 3: _____ COST: _____	
Advantages	Disadvantages

c) Identify the points of difference between the suppliers

SUPPLIER 1: _____	
Points of Difference	
SUPPLIER 2: _____	
Points of Difference	
SUPPLIER 3: _____	
Points of Difference	

d) Make a decision as to which supplier you will use, providing reasoning for your decision.

2.4 Investigate user-friendliness, system support and training issues

Introduction

Now that you have conducted research into the various asset options and advantages and disadvantages of each supplier, as part of the decision making process it is also important to identify or review the key purchasing priorities that will guide the purchase.

In Section 2.6 of this manual, the importance of budgets and purchasing within budgets will be discussed in more detail. Therefore for the purpose of this section we will focus on identifying the non-budget related purchasing priorities.

Identify key purchasing priorities

Naturally all computer asset purchases need to consider budget implications.

Whilst it is essential that costs meet a budgetary figure, there are other purchasing priorities that need to be considered. Frequently, the amount actually spent is not the central element in determining whether or not the expenditure is deemed necessary and appropriate.

Following are examples of key purchasing priorities that will guide the non-financial decision making process. These examples include, but not limited to:

- Ease of use of the asset
- Ease of integration into the existing workplace – in terms of other equipment, systems, etc.
- Reliability of the item – freedom from breakdowns, low-level of service/maintenance costs
- Safety – no injuries relating to use of the item
- Compliance – full compliance with all legally imposed obligations
- Asset achieves or exceeds the results/outcomes expected of it
- Resale value
- Enhanced service levels
- Support services
- Training requirements and training provided by suppliers.



Knowing what your workplace identifies as the contextual elements for determining what is realistic expenditure functions to also identify the factors that you need to consider before committing to the final purchase.

If ease of use is a high priority, then you need to make sure that whatever you buy can be readily learned and easily used. In many ways, realistic expenditure is a reflection of the value-for-money that a purchase represents to the business as defined by the relevant stakeholders.



Activity 12 - Identify key purchasing priorities

Discuss the key purchasing requirements, excluding budget, that would guide the decision making purchase of computer assets for your tourism organisation.



budget, that system

What are the factors / priorities?	What weighting/priority does each of them have?
Ease of use of the asset	
Ease of integration into the existing workplace	
Reliability of the item	
Safety	
Compliance	
Asset achieves or exceeds the results/outcomes expected of it	
Resale value	
Cost savings	
Enhanced service levels	
Support services	
Training requirements and training provided by suppliers	
Other factors / priorities	

Are there certain non-negotiable elements that have to be present regardless of all else?

Importance of computer hardware and software support services

Given the nature of the tourism industry where many businesses operate 7 days a week there are likely to be times when in-house computer support is unavailable.

In addition, there is always the possibility that there may be situations or problems that even experienced operators are unable to address, solve or rectify.

In these cases, some degree of technical support and services must be sought.

This may be provided by suppliers or through other technical support companies.



Support services

On-line help

On-line help desks normally provide a comprehensive solution for immediate IT, hardware and software issue resolution needs.

Suppliers or other specialists may provide remote access software and a team of dedicated help desk engineers designed to minimise downtime and provide assistance to a tourism organisation.

If the computer system is connected to the Internet, on-line help can be very effective.

Providers of this service charge a fee for you to contact them and receive on-line assistance.

The fee can be on a fee-for-service basis, meaning you pay every time you make contact and ask for help, or your establishment can enter into a 12-month contract where you pay \$X per computer, and have an unlimited access to the help.

Some providers may have a 'fair use' clause in the contract limiting the number of calls per unit.

Naturally, the precise nature of each contract or fee-for-service call varies between providers of this service.

Suppliers may offer:

- Basic level support – such as phone support during nominated, restricted or business hours
- Extended or advanced level support – such as a dedicated phone number where help can be obtained 24/7
- On-line support – this is where the system supplier undertakes to respond by email or by phone to your query within a nominated period, commonly 24 hours.

E-mail access to this service normally carries some guarantee of response times to calls. This can be 'within 8 hours', or 'as soon as possible'. This means that when you e-mail your provider saying you have a problem, they will respond to your query within this timeframe.

Telephone contact usually produces an immediate response, waiting times and queues notwithstanding.



Accessing on-line help means that the service provider can 'enter' and investigate your computer over the 'Net, using diagnostic tools, identify the problem and remedy it.

These facilities are usually available 24 hours a day, 7 days a week.

They can assist in solving hardware and software problems as well as system conflicts.

While these services are potentially cost effective, in the majority of cases costing substantially less than an in-house service call, there is no guarantee that the problem can be rectified.

Many also do not provide warranties for help provided. Some, though, give full refunds if the problem can't be fixed.

In addition, some on-line help make it clear in their policy that they will not be held liable for lost data, delays or errors resulting from their help.

Most also state in their terms and conditions that they can't be held liable for any damage to hardware arising from their advice.

Manuals and training booklets

These may accompany the hardware and software that is to be purchased.

They can be used as a total learning package. This may often be supplemented by one-to-one or group training delivered by the supplier, either in the workplace on or with the new equipment, or in a head office on replica hardware and software.

The manual may be produced by licensed, certified or accredited enterprises, or it may be a more simplified and general offering from a training provider.

Most manuals, booklets and texts arrange their content in some logical form. This can be identified and followed with reference to a 'Contents' page, or the index at the rear of the publication.



Training

Some companies and suppliers will provide a range of training services to help staff become familiar with new hardware and software assets.

This training may be:

- Initial training - when new equipment is purchased and installed
- Ongoing training - to reinforce knowledge and skills
- Updating training - conducted when new pieces of equipment, upgrades or new applications are implemented.





Activity 13 - Identify support and services

You are required to undertake research into possible support services available in your region.



Name of supplier/company	Types of support provided	Costing structure

2.5 Seek external specialist assistance as required

Introduction

Naturally the planning, design and implementation of computer systems is a complex initiative.

No one manager will have the necessary skills and knowledge to undertake this exercise alone.

They will require the assistance of various specialists for a variety of reasons.

In this section we will explore the different types of general assistance and specialist IT services that may be called upon to help make the selection process easier but also to enable a smooth implementation of assets purchased.

Types of general assistance

The following are types of general assistance that can be called upon when making the decision regarding computer assets.

These can include, but not limited to:

- Your accountant
- Your auditor
- Owners of the business
- Managers
- Supervisors
- Staff, teams and workgroups
- Customers
- Suppliers
- IT consultants.



IT specialist services

Besides the general types of assistance mentioned, there are specialist services that can be provided to help with any aspect of computer system operations.

Following are a number of examples of specialist services that may be available.

Network design and installation

These services include, but not limited to:

- Initial strategic planning
- Applications development
- Systems integration
- Expand their existing computer systems to integrated, cross-network, multi-database and accounting systems
- Design and install all sizes of networks



- Ensure connectivity with virtually all types of systems
- Make recommendations on all aspects of computing, both pre and post installation, covering such topics as cabling, type of hardware, bespoke software, operating systems, security, communications, disaster recovery and scalability
- IT upgrades, installation or migration.

IT Consultancy

Support service providers may also provide services to help tourism organisations use emerging IT technologies to help support their business endeavours.

These solutions may include, but not limited to:

- Remote, onsite and 24/7 services
- Remote working away from the office with a virtual desktop
- Mobile working for emails on the move
- Email security, anti-virus, anti-SPAM solution
- Secure offsite backup of data
- Bespoke Software Application Development
- Website and e-Business design.



Software Application Development

Support specialists may also undertake application design; development and integration services create custom business computer system applications.

The services may vary from initial requirements definition, to architecture design, application selection and integration, and development and deployment.

Services may include:

- Customisation and integration of third-party software
- Design and development of content management systems for corporate intranets and extranets
- Internationalisation and localisation of Web applications.



Activity 14 - Identify general assistance and specialised IT services

You are required to undertake research into possible general assistance and specialist services available in your region.

Name of supplier/company	Types of general assistance or specialist services provided	Costing structure

2.6 Make decisions on computer acquisitions in consultation with appropriate colleagues

Introduction

To date, we have looked at a number of aspects that need to be considered when purchasing new computer system hardware and software assets.

From undertaking detailed identification, researching different options and suppliers, identifying key purchasing priorities through to identifying support and specialised services that are used to establish and maintain business systems, we now have a good understanding of what is required and what is the best solution to meet these needs.

In this section, we will explore in more detail the activities associated with making final purchasing decisions.

In essence, it will focus on the financial aspects relating to the purchase.

Consider budget requirements

Naturally the greatest influence in purchasing computer assets will be the budget.

The computer assets that a business can obtain are often limited by the budget available for purchasing items.

Management will determine the level of funding available for computer system assets and make an amount available through its budgets. The budgets may be a set dollar figure or a percentage of revenue.

There is no doubt that budget and cash flow concerns are one of the biggest obstacles to obtaining required resources.

Later in this section we will look at the different purchasing options available.



Activity 15 - Impact of budgets

Discuss the impact of budget on obtaining computer assets for the business.

Making efficient use of budget resources

Using budget resources efficiently can be construed in different ways and where there is an expectation that you make efficient use of budget resources you must clarify what this phrase means to those who are going to sit in judgement on your actions.

To some, efficient use of budget resources simply means acquiring an item within given budget restrictions. If you come in under budget, you have been efficient.

To others, efficient use of budget resources can involve consideration of:

- The total cost allied to the purchase of a computer asset
- The effectiveness of the computer asset in the actual workplace situation – looking at how well the item performs the job it was bought to do
- The extent to which the computer asset can be used for more than one job – so that cost can be spread over different budget lines and so that the maximum use can be made of the item for the period of its working life.



Undertake a cost–benefit analysis

An accepted way of ensuring realistic computer asset expenditure is to undertake a cost–benefit analysis in relation to the asset that is to be purchased.

A cost–benefit analysis is a common requirement in many organisations for all expenditure over a stated amount.



The cost–benefit analysis considers the costs and related benefits associated with purchasing computer assets and allows a mechanism where various alternatives can be compared against each other with a view to identifying the single most beneficial purchasing option for the business.

The cost–benefit analysis is a wide-ranging analysis that takes into account tangible and intangible factors associated with the purchase, and attaches a dollar value to each identified tangible and intangible cost and benefit.

Identify all costs

When considering the costs associated with a purchase, issues that may require attention can include:

- Purchase price – including all parts and add-ons, options, taxes, delivery fees
- Opportunity cost – this is the cost to the business of spending money in one area meaning it can't spend that money in another area (the opportunity to do so is lost once the dollars have been spent)
- Cost of finance, where applicable – including fees and charges, interest rates, penalties for late payment
- Testing or compliance costs – it would be foolish to purchase equipment that was not tested before it was shipped to the buyer

- Transportation costs – equipment which is usually large in size may limit transportation options and even require special rigging to secure safety and security during transportation
- Site preparation costs – the site must be prepared for incoming equipment including alterations to infrastructure and office design
- Installation costs – is installation included in the cost of the equipment?
- Make-ready costs – installation and pilot runs that go beyond installation costs. Period testing plus minor to major adjustments
- Tooling, parts and suppliers – although considered as a secondary expense, these items can be costly and not part of the initial costs of maintenance
- Spare parts costs – what is the cost associated with spare parts, are they available, lead time for them and is that cost included in the purchase price?
- Training costs – replaced equipment will require minimal operator training. New equipment may require training the operators fully on maintenance and breakdown
- Cost of power and consumables to keep the item operational
- Service and maintenance costs
- Insurance costs for the new item
- Cost of adapting existing systems and procedures to accommodate the new purchase
- Lost sales caused by interruption to normal business
- Reduced levels of earnings while staff learn the new item
- Depreciation.



Identify all benefits

Benefits relating to the purchase of a new item may relate to:

- Higher levels of earnings/revenue – which can relate to increased speed of servicing customers
- Reduced levels of unsatisfactory services
- Energy savings – brought about by advanced technology
- Labour savings – obtained where the item increases productivity
- Savings on raw materials – where the item makes more efficient use of base products
- Sale value of any equipment that is being disposed of to make way for the new item
- Enhanced image or market positioning for the business by virtue of having the new equipment.



The dollar values attached to costs and benefits will either be:

- Fact/evidence-based – calculated on known statistics, costs, quotes, historical records, advice from suppliers
- Estimates – based on an intimate knowledge of the industry, the business and the item in question.

The keys to any effective cost–benefit analysis are:

- All costs must be identified and quantified in dollar terms
- All benefits must be identified and quantified in dollar terms
- All purchasing options must be subjected to a thorough cost–benefit analysis
- All cost–benefit analyses must be compared to determine the most effective option.

It is important to note there are sometimes situations where, despite extensive cost–benefit analysis and despite all relevant research and investigation, a purchasing decision is made based on some other criterion.

For example, the owner of a business may simply have a personal preference for a certain supplier, brand or option and no end of statistics, evidence, etc. will change their mind.



Activity 16 - Conduct a cost-benefit analysis

Based on the computer system required for your organisation, you are required to undertake a cost-analysis.

IDENTIFY COSTS

Please identify possible types of costs, including an approximate monetary value for the cost.

You may use an excel spreadsheet to detail all the costs relating to the purchase of a computer system.



benefit

Type of cost	Monetary value for the cost

IDENTIFY BENEFITS

Please identify all the possible benefits associated with the purchase of a new computer system.

Type of benefit	Monetary value of the benefit



Identify methods of financing

When purchasing computer system assets, it is important to identify the most efficient use of finances.

In summary, the different methods of financing include

- Purchasing
- Leasing
- Hiring.



Purchasing

Purchasing makes sense when the asset is:

- Integral to the organisation - building extension
- Has long term depreciation - has useful life of 25 years
- Has capacity for a variety of profitable uses - areas or equipment that you use and rent out
- Has an ultimate salvage value - someone will buy it.

Many organisations have purchased expensive technology as purchase was seen as security, only to find it depreciating badly, in the real sense of salvage value, unproductive due to technical hitches cropping up semi-regularly, spare parts costing a fortune, unresponsive markets, cheaper and better competition down the road, rises in interest rates on the loan taken out to purchase, and a resultant bitterness with the 'investment'.

The recurrent budget must be able to support the costs of the capital investment such as loan servicing, amortisation, maintenance cost. On the other hand the investment could have been a superb market move that has excellent ROI.

Pros and Cons of purchasing

Pros

- Excellent if paying cash upfront is possible
- You own the equipment.

Cons

- Can reduce cash flow
- Equipment may become obsolete faster than anticipated
- If under a limited warranty, may find repairs or maintenance expensive after warranty expires.



Leasing

A lease is a contract by which one party, known as the owner or lessor, transfers to another party, known as the lessee, the right to possess and use property owned by the former for a fixed period of time, in exchange for either a lump sum or more usually periodic instalments.

Put simply, it is property that is rented by contract for a certain period of time.

Leasing effectively means you can obtain capital assets but use the recurrent budget for servicing them. You do not own the equipment; you pay for its useful life progressively to the owner, via a financier, and hopefully obtain enough ROI on the asset that exceeds lease costs.



At the end of the lease you may be offered an option to purchase the asset at the residual value, or to renew the lease with a new version of the asset.

Leasing makes sense when the item is capital intensive and expensive, technically complex requiring specific servicing or liable to have a life span that will be kept short by the market, technology, usage or its own complexity.

Types of Leases

Finance Lease

This is a method by which a company eventually gains ownership of the equipment. The Lessee retains ownership of the equipment during the term of the lease, and makes periodic payments to a finance organisation.

The Residual Value is the estimated market price of the equipment at the end of the lease, and is a fixed amount, written into the lease as part of the lessee's financial commitment.

This type of lease is classified as an open ended lease.

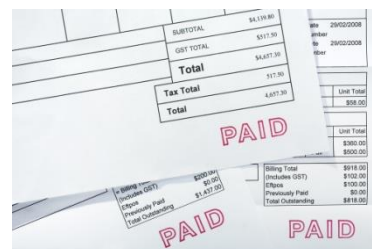
Operating Lease

- Are like rental agreements and are not equipment purchase transactions
- Usually involves the goods being leased for part of their estimated life
- There is rarely any residual value and the lessee does not get the opportunity to take ownership of the equipment – regarded as a close ended lease.

Pros and Cons of Leasing

Pros

- Cash flow regulated
- No credit tied up
- Upgrades are easier
- Tax deductible
- Service plans included
- A number of different types of lease available.



Cons

- More expensive than buying
- In some leases, equipment is returned to the lessor at end of lease
- Short lease terms yield higher payments
- Penalties for cancellations
- Service plans may last only for first year of lease.

Hire-purchase

This is a foot in both camps. Ownership doesn't transfer to you until completion of a series of time payments but you have the asset and its ownership is redeemed to you at the end of the payment periods or number of payments. Unless you have a very good reason to want to keep that particular asset, and a loan to the facility is out of the question, then it is not recommended. There's too much to lose, too little to gain.

Hire-rental

Hire rental offers the user the use of equipment on a short or long term basis. The costs are normal business expenses and are not subject to depreciation. One benefit of hire rental is that you can return the item at any time without incurring extra costs. Hire rental is an excellent choice for emergency situations, short term high wage needs and flexibility.

Specialist equipment is available. Maintenance is carried out by the rental company.



Activity 17 - Identify methods of financing

Based on the computer system required for your organisation, you are required to identify the financing for the computer system.

Please identify possible types of costs, including approximate monetary value for the cost.



tourism
method of

an

Item	Cost value	Method of financing

Making a final financial decision

Ultimately, the financing decision for a tourism organisation is based on its balance sheet.

More important than the acquisition method are the following factors:

- Will the asset significantly improve customer satisfaction?
- Will the asset rectify a problem or issue that is substantive in nature?
- Is acquisition desirable or essential?
- What are others in the industry doing?
- What is cash flow and overall liquidity like?
- What investments exist to reinforce a decision?
- How many “volatile” factors exist to affect the asset’s performance and potential ROI? - a ‘what-if’ analysis from good to bad
- What is the ROI of the asset as a % of the facility’s turnover?
- What is the asset’s percentage internal rate of return?
- What is its pay-back period estimated to be? (purchase)
- What is its ROI relative to loan payments? (purchase)
- What is its ROI relative to lease payments? (lease)



One effective way of measuring purchase against lease is the internal rate of return, which is the effective percentage annual cost of leasing versus outright purchase, with loan servicing included as well as respective maintenance arrangements for purchase and lease.

Evaluation techniques

Property managers are often required to purchase plant, equipment, furniture and fittings. To do this effectively they use three different techniques to evaluate various options.

Payback period

This is the length of time taken to recoup original investment. The savings made by using the new equipment means eventually it pays for itself. This is referred to as a non-discounted cash flow technique.

Example

A tourism establishment wishes to purchase a new office computer system with an initial outlay of \$200,000. It is estimated that the new system will provide cash savings every year of \$35,000 due to greater efficiency and reduced staff operating time. The system has a useful life of 10 years.

Payback period = Original Investment / Average Annual Cash Flow

= \$200,000 / \$35,000

= 5.72 years

This would need to be compared with management's benchmark for projects, which may require projects to be recouped within 10 years. If this is the case the cost of the project would be accepted.

1. Discounted cash flows

The other two techniques are discounted cash flow techniques. The method basically asks – What will \$1.00 be worth in x years at a predetermined interest rate. The interest rate is usually the cost of financing the project. The interest rate table to be used in this unit is:

Table 1 - Interest Tables

Present Value of \$1

(Assuming that the \$1 is received in a single payment on the last day of the year)

Years	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1	.9524	.9434	.9346	.9259	.9174	.9091	.9009	.8929	.8850	.8772	.8696	.8621	.8547	.8475	.8403	.8333
2	.9070	.8900	.8734	.8573	.8417	.8264	.8116	.7972	.7831	.7695	.7561	.7432	.7305	.7182	.7062	.6944
3	.8638	.8396	.8163	.7938	.7722	.7513	.7312	.7118	.6931	.6750	.6575	.6407	.6244	.6086	.5934	.5787
4	.8227	.7921	.7629	.7350	.7084	.6830	.6587	.6355	.6133	.5921	.5718	.5523	.5337	.5158	.4987	.4823
5	.7835	.7473	.7130	.6806	.6499	.6209	.5935	.5674	.5428	.5194	.4972	.4761	.4561	.4371	.4191	.4019
6	.7462	.7050	.6663	.6302	.5963	.5645	.5346	.5066	.4803	.4556	.4323	.4104	.3898	.3704	.3521	.3349
7	.7107	.6651	.6228	.5835	.5470	.5132	.4817	.4524	.4251	.3996	.3759	.3538	.3332	.3139	.2959	.2791
8	.6767	.6274	.5820	.5403	.5019	.4665	.4339	.4039	.3762	.3506	.3269	.3050	.2848	.2660	.2487	.2326
9	.6446	.5919	.5439	.5003	.4604	.4241	.3909	.3606	.3329	.3075	.2843	.2630	.2434	.2255	.2090	.1938
10	.6139	.5584	.5084	.4632	.4224	.3855	.3522	.3220	.2946	.2697	.2472	.2267	.2080	.1911	.1756	.1615

2. Net Present Value (NPV)

We have already established that by purchasing the new computer system for \$200,000 we will make savings of \$35,000 per annum. The interest table is used to re-value the \$35,000 for each of the future 10 years. This is because \$35,000 in today's terms will equal a lesser and reducing amount in each of the following years.

Example: Using the present value tables above and a 12% discount rate:

Note that the last two columns are positives.

Year	Cashflow (A)	Present Vale (PV) @ 12% (B)	(A) x (B)
0	-200,000		
1	+35,000	x .8929	31,251
2	+35,000	x .7972	27,902
3	+35,000	x .7118	24,913
4	+35,000	x .6355	22,242
5	+35,000	x .5674	19,859
6	+35,000	x .5066	17,731
7	+35,000	x .4523	15,831
8	+35,000	x .4039	14,137
9	+35,000	x .3606	12,621
10	+35,000	x .3220	11,270
		Total	197,757

In year 0, only the cash outflow for the purchase is shown (-200,000) in (A).

Using the present value tables, in the next row (Year 1) we have shown the savings of \$35,000 for the year (A), then the rate to be used for discounting at 12% (B). The next column shows the result of multiplying column A by column B, which for year 1 means that \$35,000 at 12% discount = \$31,251

We are expressing tomorrow's money in today's value because money usually decreases in real value.

For the above example the value of all the cash inflows from the project discounted @ 12% is \$197,757, whilst the cash outflow at the beginning was \$200,000.

Therefore the NPV of the project is \$197,757 – \$200,000 = -\$2,243.

Because the NPV is negative the project would not be accepted at a rate of 12%.

If you do the above example at a lower discount rate a positive NPV would occur resulting in the project being accepted.

3. Internal Rate of Return (IRR)

The IRR is calculated by discounting the cash outflows of the project until the Net Present Value (NPV) of the cash inflows equals zero or a small positive result. This in turn shows the rate that can be compared to the market place interest rates. It uses trial and error to determine the correct discount rate that will bring the cash outflows and inflows into balance.

Management must then decide if that rate is realistic and if so they will proceed with the purchase.

In the following example, using the same data as above, we have shown the discounted values when using 10% and 8%.

Example:

Year	Cashflow – (for outflow) (A)	PV rate @ 8% (B)	PV rate @ 10% (C)	(A) x (B) i.e. 8%	(A) x (C) i.e. 10%
0	-200,000				
1	+35,000	x .9259	x .9091	32,407	31,819
2	+35,000	X .8573	x .8264	30,006	28,924
3	+35,000	X .7938	x .7513	27,783	26,296
4	+35,000	X .7350	x .6830	25,725	23,905
5	+35,000	X .6806	x .6209	23,821	21,731
6	+35,000	X .6302	x .5645	22,057	19,758
7	+35,000	X .5835	x .5132	20,423	17,962
8	+35,000	X .5403	x .4665	18,911	16,327
9	+35,000	X .5003	x .4241	17,511	14,843
10	+35,000	X .4632	x .3855	16,212	13,492
			Totals	234,856	215,057

The total cash inflows at 8% are \$234,856 and at 10% is \$215,057. Therefore both rates would be acceptable to management as they return higher than the initial cash outflow of \$200,000. Probably the NPV rate to achieve zero or close thereto would be about 11.75%

Remember that these are speculative rates, and are not necessarily true and should not be relied upon without further supporting evidence that the purchase is beneficial to the organisation.



Activity 18 - Evaluate the purchase of a new computer system

Using the three techniques discussed, evaluate the purchase of a new computer system based on the following information.

- The system will cost \$175,000
- It has a useful life of 8 years
- It will generate a saving of \$25,000 per annum
- The establishment requires all projects to have a discount rate of 7%. (Refer to Table 1)
- The establishment requires a payback period of less than 5 years.



Questions

1. Calculate the payback period. Based on the criteria above, should the purchase proceed?

2. Calculate the IRR based on the 7% discount rate. Will the purchase be acceptable to the establishment?

3. Calculate the NPV of the project and estimate. Is this acceptable to the establishment?

Identify necessary prerequisites prior to purchasing assets

The following are critical prerequisites to acquiring computer system assets:

- To know what the item is required to do – which can include knowledge about who will use it, how many will use it, where it is to be physically located, limitations that apply to size, weight, noise emissions, vibration, etc. In the broadest sense, this demands that you know all the applicable operational parameters for the item
- To know the businesses with whom your organisation will deal and those with whom it will not do business – this embraces preferred suppliers but also individual organisations with which there may have been unacceptable previous transactions
- To know the preferred method of acquisition for the business– including knowing how much money is available for the acquisition. This can include knowing scheduled repayment amounts for leasing and similar arrangements, the amount of deposit immediately available to secure an item, the all-up cash available to make a cash purchase
- Implicit in this is also a need to know:
 - The terms that your business wants applied to the purchase – such as when first payment is due, the amount of credit made available by the supplier
 - Non-negotiable elements of any deal – which may include delivery and installation dates, need for the supplier to provide on-site training, insistence on a guarantee that service and parts will be available for X years
- To have the necessary authority to go ahead and make the purchase – this can include formal authority to purchase via a purchase order from the business, performing on-line banking transactions or ensuring that online arrangements have been made identifying you as able to lodge an online order on behalf of your company.



2.7 Plan introduction of new computer systems to take account of impacts on colleagues and customers

3.5 Set up equipment to accept the software and functions being used

Introduction

Now that all computer system purchasing decisions have been made, it is now time to prepare for its implementation.

Naturally the greater the degree of change to hardware and software will have a greater impact on staff, ranging from greater anxiety to lower productivity.

Over time the implementation of the system assets will make the job of tourism staff easier, however it is important that not only management understand the short term concerns but also to let staff know they acknowledge these concerns and are there to help them.



Planning for the implementation of computer system assets is essential to ensure all aspects relating to the proposed actions have been considered, planned and have clear plans for implementation.

This section will explore some of the activities associated with the implementation of computer system.

Implementation considerations

When implementing new computer system assets there are a number of considerations including but not limited to:

- Identify all implementation actions to be taken
- Allocation of responsibilities for action to nominated persons
- Determination of accountability for actions taken
- Timelines for completion of nominated actions
- Compliance requirements
- Occupational health and safety (OHS)
- Arranging arrival and security of new computer assets
- Timing of new system assets to be implemented
- Saving information from old system assets
- Removal, storage and sale of old computer assets
- Arranging location and physical layout of new assets
- Unpacking and installing new computer assets



- Timing issues
- Resources
- Aligning new computer assets to existing structures, hardware and software systems
- Integration with other aspects of the business
- Adding information to new computer system assets
- Changes to policies and procedures
- Documentation & record keeping
- Impact on organisational culture
- Anticipation of problems
- Training for staff to become knowledgeable and productive with new computer system assets.



Communicate implementation actions

Like with any change, it is vital that everyone who will be impacted by the introduction of a new computer system understands what is taking place as part of the implementation, how it will affect them and what they need to do.

It is important that everyone knows what is expected of them. In these times of uncertainty everyone must be aware of the roles each person plays.

As mentioned, the greater the degree of change to the computer system, the greater the need to clearly communicate roles and responsibilities.

If only small asset changes are taking place, this process may be very minimal with little change or impact on staff.

This is essential to ensure that key aspects of the implementation are not only communicated with all concerned but also understood by them.

These aspects can therefore include:

- Reasoning or purpose behind the changes to the computer system
- Identification of the key changes that are taking place
- Identification of how the new computer system will benefit them
- Start dates of implementation
- Expected completion dates
- How the implementation will be implemented
- Activities to be performed
- Resources required to implement the plan
- Work breakdown
- Involvement of other staff
- Roles and responsibilities of staff
- Pre-change activities including attending meetings, reading job descriptions etc.



- Timelines for implementation
- Training activities that will take place
- Support mechanisms
- Expected standard of performance
- Milestones for completion
- Budgets to work within
- Methods of communication
- Making observations and providing feedback.



Communicating implementation actions

The communication of new computer system actions may occur in various ways.

The key to communicating these changes is the communication must be planned.

Your approach should follow these guidelines:

- Hold a formal staff meeting to inform stakeholders of the implementation – if they have been actively involved in the planning process then the concept should not come as anything new to them
- Explain the new computer system assets to be introduced
- Give them a hard copy of materials to help explain and use new computer system assets including:
 - Brochures and specification sheets
 - Operational manuals
 - Copies of standards
 - Copies of policies
 - Copies of procedures
- Explain the changes – tell them how the 'new' system differs from existing operations
- Encourage questions about the implementation plan – answer them fully and honestly
- Make yourself available outside this meeting to talk to if they have concerns regarding the operational plan
- Include new policies, procedures etc. in operational manuals and in-house training.





Activity 19 - Communicating computer system implementation plan

Briefly describe how you will communicate the implementation of a new computer system to staff in your tourism operation.

Staff / stakeholders	Information to be communicated	Method used to communicate information

Accompanying documents and items

When computer system assets are purchased, it is usual for them to be accompanied by documents and items that must also be taken care of.

Every legitimate software program sold to a property will come complete with a comprehensive set of instructions. There may be just one copy, or there may be multiple copies.

Types of documents

Documents that may accompany the delivery of assets can include:

- Guarantees and warranties
- Installation instructions
- Wiring diagrams
- Manuals
- Trouble-shooting guides
- Spare parts lists
- Maintenance schedules.



A file should be established to store these documents and the unpacking, installation; delivery process must ensure these are not inadvertently discarded. You may also elect to include photocopies of purchase orders, receipts and invoices.

Using manufacturer guidelines

This is a pretty straightforward area.

Firstly, make sure you keep any and all manufacturer instructions for use and future reference.

Secondly it is essential to read the booklets, brochures or other relevant documents

Instructions and guidelines are printed to assist in the operation of the relevant equipment: they are based on tried and tested practical applications, so they work. It doesn't make any sense at all to ignore them or to try something different.

In some cases, too, not adhering to the guidelines or instructions may void warranties.

The guidelines or instructions will refer not only to operational matters and troubleshooting, but also to preventative maintenance techniques.

Types of information contained within documents

You must obtain a copy of this manual, guide or instructions and read it to become familiar with:

- The conditions of use of the system and program
- The capabilities of the system
- How to use the system
- Trouble-shooting techniques
- How to access support
- Training materials – such as CDs or specific training sites which are not integrated into actual operational areas.

Identify predictable problems with implementation of new computer system assets

As implementation of new computer system assets takes place, things often do not go exactly to plan for a variety of reasons.

Whilst on paper, the transition may seem easy, when the actual computer assets are implemented and integrated with existing systems, problems may occur.

In addition, whilst new systems may appear easy to use, this may not be the case as staff become familiar with new systems.

Quite often as the implementation takes place, further change or correction is required.

As mentioned, any change to computer systems brings with it a certain level of unpredictability.

Managers must understand this and be realistic in how operations will be disturbed and what productivity levels can be achieved.

Whilst managers will not be able to predict all problems associated with the implementation of new computer systems, some predictable problems include, but not limited to:

- Delays in implementation
- Frustrations with new systems
- The physical space or area allocated to new computer assets is inappropriate, insufficient or does not fit
- Lack of seamless integration between old and new systems
- Greater stress as workload increases and productivity reduces
- Staff are unfamiliar with new systems
- Staff unable to perform functions on new systems
- Staff making mistakes in new system
- Greater complaints due to delays and inaccurate information
- Lost data
- Limited access due to new security of password procedures
- Disruptions to routine.





Activity 20 - Anticipate and resolve problems associated with implementation of new computer systems

Briefly describe how you will communicate the implementation of a new computer system to staff in your tourism operation.

Types of anticipated problems	Actions to resolve problems
Delays in implementation	
Frustrations with new systems	
The physical space or area allocated to new computer assets is inappropriate, insufficient or does not fit	
Lack of seamless integration between old and new systems	
Greater stress as workload increases and productivity reduces	
Staff are unfamiliar with new systems	
Staff unable to perform functions on new systems	
Staff making mistakes in new system	
Greater complaints due to delays and inaccurate information	
Lost data	
Limited access due to new security of password procedures	
Disruptions to routine	
Problems with productivity	
Extra support required by employees	

Other anticipated problems

Provide support and resources to support implementation actions

As mentioned, any time change takes place, whilst in the long term it will benefit the organisation and staff involved, it will cause disruptions and make activities harder for staff in the short term.

This is evident not only as the actual change takes place, but as staff try to understand and become familiar with the new way of doing things.

It is especially important that during this stressful time, that management play an active role in leading and supporting the team as a whole and individuals who may need assistance.

There are many ways a manager can provide assistance and support, however at the end of the day, the aim is to enable staff to be able to learn and implement new changes into the workplace.

During this period, you must manage the twin functions of 'group task' and 'group maintenance'.

Task functions

'Task' functions of leadership involve leading the group successfully understanding and performing new activities associated with change. Factors included here are:

- Setting plans to help staff achieve the set change goals
- Passing on facts and skills
- Offering ideas and information
- Seeking opinions
- Giving directions
- Getting the individuals within a group to function as a cohesive unit
- Coordinating activities
- Clarifying goals as they progress through the change process.



Maintenance functions

'Maintenance' functions centre on ensuring that staff continue to work harmoniously and that there are good working relationships among the team. This includes your being involved in:

- Provision of positive feedback to staff – to help keep them interested, engaged and motivated
- Giving encouragement – either verbally or non-verbally
- Raising enthusiasm amongst the team – to make them aware that they can achieve what is required, and to provide motivation during the tough times when they might be tired, frustrated or disinterested
- Maintaining a safe, secure and supportive physical and emotional environment – physical safety is, of course, important, but so, too, is the need to provide emotional support when staff find the going tough or when they feel dispirited
- Acknowledging contributions and efforts made by staff – the important thing is to provide acknowledgement for effort and not just for success
- Using humour as a motivator and means of reducing tension – this can assist in maintaining momentum and getting staff to have another go at something they may have previously not succeeded in
- Sensitive and non-judgemental communication – it is very important that whatever is said to staff cannot be misconstrued in any way. Your role is to support, encourage, nurture – not to criticise or put people down
- Frequent, accurate and encouraging feedback – which means finding something to provide a positive comment about even when staff are unable to achieve the set objective
- Allowing staff to make mistakes and create an environment where it's okay to do so.



2.8 Ensure staff are adequately trained in using new systems

1.4 Identify computer training needs and provide training opportunities

Introduction

Before staff are required to use new computer systems it is vital that all staff competent in not only the functions associated with the system but also in being able to access and interpret information which is contained within.

As can be seen when undertaking any activity on a computer system being able to locate, identify, interpret, use or provide information to tailor and improve guest service in a timely manner is essential.



In essence for staff must be proficient in the use of a computer system requires two things:

- An understanding of what can be done, why it is done, and when it should be done
- Practice in applying that knowledge and understanding to the system.

In this section we will explore the different types of activities and actions that may be required by tourism staff when using new computer systems and the associated software and programs that they use as part of their normal duties.

General computer system knowledge and skills training

Before staff can perform specific skills they must have general knowledge and skills to operate a computer system and its associated hardware and software.

The staff must be trained in:

- Basic computer and keyboarding skills
- Literacy skills to read and interpret complex product information controlled by the system which can include costs, terms and conditions of their sale
- Read, interpret and use system codes
- Writing skills to input bookings or operational data accurately
- Numeracy skills to interpret statistical data within the various reports produced and manage the accounting processes that relate to a particular file, customer or booking
- Organisation's policies, plans and procedures, especially in regard to file-naming and storage conventions
- Correct log-on and shut-down procedures for computer equipment
- Organisational IT procedures including back-up and virus protection procedures
- Role of computer and operations systems within the tourism and hospitality industry
- Range of products and services controlled by the computer system



- Range of sales, operational and accounting functions that can be controlled by the system
- Common computerised reservation and operational entries, including encodes and decodes
- Mandatory fields
- Applying access codes
- Interpreting and using on-screen prompts to determine required information – these may include prompts such as ‘Yes’, ‘No’, ‘OK’, ‘Continue?’, ‘Apply’, ‘Update?’, ‘Edit’ and ‘Print’
- Interpreting and applying on-screen abbreviations, acronyms and options
- Using navigation tools such as buttons and tabs
- Applying correct date formats within the system
- Adhering to system protocols and field size limitations
- Completing required fields
- Describing the field and menu links available between screens
- Responding to system-related queries
- Accepting, confirming, denying, amending or following-up bookings
- Forwarding sub-menu bookings - such as vehicle hire, theatre tickets, local tours, entry to local tourist attractions etc. to the appropriate individual or business
- Translating communications from/into other languages
- Converting currencies – quotes, rates etc.
- Integrating updates from suppliers such as product descriptions, terms and conditions, prices, packages etc. into current property information fields so that we maintain currency of information
- Negotiating new deals based on new information or emerging trends for listing
- Adding distribution channels to the CRS network – such as new agents etc.
- Generating invoices, accounts, folios and statements and distributing them as required
- Generating and distributing relevant reports to other system users, affiliates, partners or related businesses – such as sales activity reports, system updates such as changes to transaction methods, protocols, and parameters, member reports, and marketing reports.
- Requirements for specific formatted entries
- Procedures for confirming, storing and retrieving reservations or operational data
- Procedures for sending and receiving messages
- Maintenance activities
- Technical terminology in relation to reading help-files and manuals
- Methods of detecting faults in and solving problems with technology.

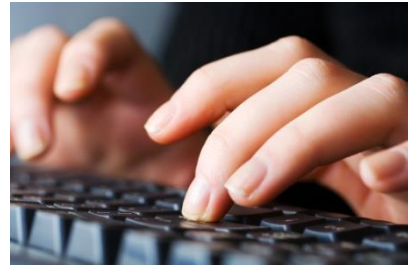


Specialised GDS and CRS training

Specialised training courses need to be undertaken to attain necessary competencies with individual systems.

These specialised courses relate to the features and functions of individual systems and address topics such as:

- Signing on and off
- Using system passwords
- Understand online system help
- System help facilities
- Key functions
- System abbreviations
- Encoding and decoding
- Displaying and interpreting availability screens
- Selling from availability screens and making direct sell reservations.
- Identifying the various levels of selling
- Creating and updating files
- Completing the Booking File
- Using service requests to generate passenger specific requirements
- Accessing and actioning queues
- Displaying fares, interpreting fare rules, and obtaining fare quotes
- Identifying and updating booking status codes.
- E-ticketing
- Accessing general information
- Completing documentation – associated with faxes, visa support letters, trip documents, creditor documents, debtor documents and client documents
- Client and Trip file management and accounting including:
 - Accounting transactions
 - Creating Transactions
 - Creating, reprinting and reversing receipts
 - Ticket requisitioning
 - Cheque requisitioning, invoices and credit notes
 - Payments and reversing payments
 - Debtors files
 - Recording fees
 - Completing documentation – associated with faxes, visa support letters, trip documents, creditor documents, debtor documents and client documents.



Types of computer system training

There are a number of ways for staff to become familiar with general and specialised computer systems knowledge and skills include, but not limited to:

- Participating in formal training which may be provided by the property or by the software supplier
- Utilisation of dedicated training software that duplicates the real operating system but which ensures the real system is not accessed, breached, modified or corrupted
- Reading the User's manual
- Observing and listening to an experienced operator
- Asking questions about what is happening and why things are done the way they are
- Writing your own notes – in some way this can be seen as your own personal Operator's manual written by you, for you to suit personal preferences and address the issues you find you have most trouble with
- On-going coaching and mentoring to enhance your competency once the basics have been mastered.





Activity 21 - Identify computer training

You are required to outline five computer courses that you would conduct to help your familiar with your new computer system.



training
staff become

Topic of training	Key points to learn	Training method

2.9 Monitor new systems for efficiency and take action accordingly

Introduction

In Section 1.2 of this manual, we looked at the importance of monitoring the performance of computer systems in determining the need for an upgrade or replacement of a computer system, either in its entirety or for specific hardware or software components.

In this section we will explore the tasks associated with monitoring the success of new computer systems after they have been implemented.

Importance of monitoring new computer systems

The accepted ways of monitoring the effectiveness of new computer systems will relate to determining:

- Achievement of key performance indicators
- The actual workplace effectiveness of the computer asset – as opposed to its anticipated usefulness
- The actual costs, savings or other benefits involved in using this asset.



Two frequent questions that the monitoring of resources seeks to answer are:

- Did we make the right decision when we bought this item?
- Would we buy this item again if the need to do so arose?

Key performance indicators

Measuring the effectiveness of business computer systems can be broken down into many aspects depending on each manager and organisation however can be broken down into the following categories:

- Satisfaction
- System usage
- Performance/ usefulness
- Productivity
- Value analysis
- Cost-benefit analysis.



Actual workplace effectiveness of computer asset

The actual workplace effectiveness of a computer asset is often the most significant determinant of whether the right decision was made, and if the same item would be bought again.

If the financial component of the monitoring process is acceptable to the organisation, then operational effectiveness is usually the key criteria for determining if the correct decision to acquire was made.

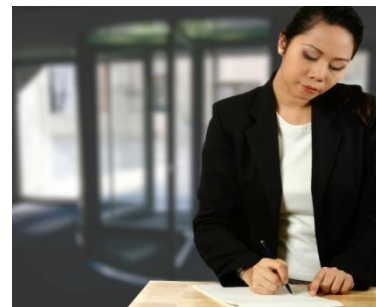
Workplace effectiveness can be judged from different viewpoints and it is appropriate that several of these be used in reaching a final conclusion.

The activities associated with monitoring the actual workplace effectiveness of computer asset include:

Personal observation

This involves you looking at the use of the computer asset in the workplace over a period of time and forming a subjective impression as to its effectiveness and contribution to the aims of the business. As you observe you should be asking yourself:

- Is it working as expected?
- Is it integrating with other systems as intended?
- Are staff or customers finding it easy to use?
- Does it appear appropriate?



Discussions with staff

These should be a combination of both formal and informal feedback from staff about the item.

- How does it perform?
- Is it easy and safe to use?
- What are the good and bad points about it?
- How does it compare to the previous item?
- Is it performing as they expected?
- What needs to be changed about it to improve it?
- What product quality does it produce?

Discussions with customers

This must occur where an item is being used to produce a product or service that is delivered directly to the customers:

- How do they feel about the product being produced? What do they like or dislike?
- How do the products or services produced by this item compare to those previously produced?
- How does it compare to similar items they have experienced being used at other similar businesses?

Discussions with other stakeholders

These may include supervisors, internal IT staff.

- What are their observations and feelings?
- How frequently has it malfunctioned, broken down, been out-of-service?
- Are replacements available when needed?
- How has it enhanced or detracted from the operation of the business?
- Has the item lived up to their expectations? To what extent and why?



Examination of the final products generated by the item

This involves assessing the product against given criteria in an attempt to incorporate an objective element into what can otherwise be a very subjective exercise:

- What proportion of products fails to meet the required standard?
- What is the nature and extent of any substandard products?
- To what extent does the product exceed the minimum expected standards?

Using other monitoring methods

There are many other tools or methods available to monitor the effectiveness of a new computer system including:

- Compliance reports
- Use of checklists
- Surveys
- Checklists
- Flowcharts
- Benchmarking.



Actual costs, savings or other benefits

The factors addressed above tend to be intangible considerations whereas this orientation shifts attention to the more tangible factors.

Essentially, this looks at statistics and asks you to investigate business records.

Areas for attention include:

- How has the company financial benefited by using the system
- Identification of the number of reported breakdowns and the down-time and lost production associated with that time
- Costs to service the item – in-line with establishment requirements or manufacturer's instructions relating to scheduled and preventative maintenance, taking into account both labour and parts
- Costs to repair and maintain the item – this identifies costs associated with rectifying problems that occurred despite the scheduled and preventative maintenance that was done

- Running costs – in terms of energy usage. This can be calculated either for the item itself or by drawing a comparison with previous items
- Checking life cycle statistics for parts – where parts of an asset needs regular replacement it is common for the supplier to provide an estimate of how long these parts can be expected to last. Was their estimation, on which you may have partly made the decision to purchase, been accurate or overly optimistic?

Ancillary considerations

Other areas that deserve consideration when monitoring efficiency of new computer system assets include:

Evaluation of the extent to which promises made by a supplier have been kept

- Was delivery and installation of the item achieved as promised?
- Did they provide the on-site training as they said they would?
- When you needed them for service, spare parts, etc. was their response in-keeping with what they said they would provide?

Evaluation of the effectiveness and cost of maintenance plans and service contracts

- Were the scheduled maintenance plans adequate and suitable to keep the item working and in operation or was there a need for other maintenance on top of what you were told was needed?
- If you signed up for a service contract for the item, has it proved to be value-for-money?
- Did the service contract prevent the item from breaking down?
- If you had cause to rely on the service contract to have the item repaired, was the service provided appropriate to your needs?
- Was the response time suitably quick?
- Did the service agent make your breakdown a priority?



Evaluation of staff training in relation to the item

- How long did it take?
- How much did it cost?
- To what extent did the need to provide staff training, or roster staff to attend staff training, impact not only on the labour budget, but also on service delivery to the customers?

Evaluation of warranties and guarantees

- If you had to rely on the warranties or guarantees relating to the items, how good did these prove to be?
- Did the supplier respond in a professional, responsible and supportive manner or did obstacles appear to be put in the way of resolving the issue?

Evaluation of the workplace usage of the item

This asks you to be fair in your analysis of the operation of the item in question and the associated costs.

You are being asked to judge how the actual usage of the item compares to the projected usage.

- Was there a much heavier demand on this item than originally anticipated that could have caused breakdowns?
- Was the item asked to do work it wasn't really designed to do?
- Did heavy usage of the item mean that required service to the item could not be performed?

Evaluation of the physical workplace location of the item

- Were there any unique situations that applied to where the item was positioned that could have impacted on its performance?
- Was it exposed to the elements?
- Was it open to interference by members of the public who could have tampered, interfered or damaged it?



Activity 22 - Identify monitoring methods

You are asked to identify different aspects that you would wish to evaluate when monitoring the performance of a new computer system.



ASPECT TO EVALUATE	METHOD OF MONITORING & EVALUATION

Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

2.1 Please complete the following activities relating to this Performance Criteria:

- Activities 6,7

2.2 Please complete the following activities relating to this Performance Criteria:

- Activities 8,9,10,11

2.3 Please complete the following activities relating to this Performance Criteria:

- Activities 8,9,10,11

2.4 Please complete the following activities relating to this Performance Criteria:

- Activities 12,13

2.5 Please complete the following activities relating to this Performance Criteria:

- Activity 14

2.6 Please complete the following activities relating to this Performance Criteria:

- Activities 15,16,17,18

2.7 Please complete the following activities relating to this Performance Criteria:

- Activities 19,20

2.8 Please complete the following activities relating to this Performance Criteria:

- Activity 21

2.9 Please complete the following activities relating to this Performance Criteria:

- Activity 22

Summary

Purchase hardware and software

Identify broad computer equipment hardware and software requirements accurately and research possible new approaches using appropriate information sources

- Sources of technology information
- Setting up a computer network / system
- Internet connectivity
- Wired and Wireless Networks
- Local area network (LAN)
- Network Setup
- Setting up a secure network
- Personal Computers (PC's)
- Monitors
- Keyboards
- Mouse
- Scanners
- Printers
- Multifunction devices (MFDs)
- Computer back-up systems
- Software packages
- Tourism computer systems
- Global Distribution System (GDS)
- Computerised reservation systems (CRS)

Assess the feasibility of acquiring or upgrading hardware and software based on available information

Evaluate options against the needs of the enterprise, including advantages and limitations

- Procurement
- Definition of feasibility
- Feasibility activities
- Methods to acquire computer system assets
- Conduct research into required computer systems assets
- The tender process

Investigate user-friendliness, system support and training issues

- Identify key purchasing priorities
- Importance of computer hardware and software support services
- Support services

Seek external specialist assistance as required

- Types of general assistance
- IT specialist services

Make decisions on computer acquisitions in consultation with appropriate colleagues

- Consider budget requirements
- Making efficient use of budget resources
- Undertake a cost–benefit analysis
- Identify methods of financing
- Purchasing
- Leasing
- Hire-purchase
- Hire-rental
- Making a final financial decision
- Evaluation techniques
- Identify necessary prerequisites prior to purchasing assets

Plan introduction of new computer systems to take account of impacts on colleagues and customers

Set up equipment to accept the software and functions being used

- Implementation considerations
- Communicate implementation actions
- Communicating implementation actions
- Accompanying documents and items
- Identify predictable problems with implementation of new computer system assets
- Provide support and resources to support implementation actions

Ensure staff are adequately trained in using new systems

Identify computer training needs and provide training opportunities

- General computer system knowledge and skills training
- Specialised GDS and CRS training
- Types of computer system training

Monitor new systems for efficiency and take action accordingly

- Importance of monitoring new computer systems
- Key performance indicators
- Actual workplace effectiveness of computer asset
- Actual costs, savings or other benefits
- Ancillary considerations

Element 3:

Maintain computer equipment and systems

3.1 Establish and monitor systems for cleaning, minor maintenance and replacement of consumables in accordance with manufacturer's instructions

Introduction

Now that new business computer systems are operational, there must be some mechanisms in place to ensure that the system and its individual components are maintained to perform at the best of their ability.

Like with any business tools, computer system assets require some type of maintenance, ranging from the simple in-house tasks to complete machine services that require specialists or technicians.

Maintenance is the process of preparing a computer asset for operation and ensuring that it is in a suitable operating condition to minimise the possible of future failure.

In most cases there is a set of stated procedures for each asset that needs to be performed on a regular basis.

It is impossible to identify all the possible procedures that all computer hardware and software assets may require, but it is important to be able to identify the various common events that will be done by employees and the procedure to be followed when outside assistance is required.

Poor or lack of maintenance on computer system assets will lead to high operating costs, which may affect the company's profit potential in the long term.

In a very competitive tourism industry, where pricing strategy is often used to compete with similar products, companies would need to ensure that the operating costs remain low in order to reap the desired profit.

While variable costs can be adjusted relatively easily, the fixed cost components like computer assets would need a very effective maintenance program to ensure that the lifespan is maximised.



Whilst the word maintenance will be commonly used in this section, in reality the process of keeping computer assets in optimal condition involves tasks relating to:

- Preventative maintenance
- Regular maintenance and servicing
- Repairs
- Physical cleaning of assets
- Cleaning of systems and internal memory
- Replacing consumables.

Role of management in maintaining computer system assets

The manager has the following responsibilities when maintaining assets:

- Planning - ensuring there is the best possible deployment of technical and human resources on maintenance tasks
- Estimating - specifying the people time and cost, plus the equipment time and cost for undertaking maintenance
- Programming - allocating maintenance work and resources at correct time intervals so that the overall facility is working at its best at most times
- Review - assessing performance and quality of work undertaken
- Corrective action - initiatives undertaken to limit future downtime and maintenance that could have reasonably been avoided.



Importance of conducting routine maintenance

While you are not expected to be a service technician, there is an expectation within the workplace that you either carry out routine maintenance on computer system assets or that you arrange for it to be carried out.

If you are unable to do the routine maintenance personally, there may be an IT person within the establishment who is charged with these duties, and it may be up to you to organise them to perform whatever routine maintenance is required.

Some premises will have a preventative maintenance program in place, meaning that the routine maintenance will be performed as a matter of course under the established schedule.



Preventive maintenance

The most effective way of ensuring computer assets are maintained well is to have a Preventive Maintenance program, also known as 'PM'. Essentially, you're detecting the need for repair before a breakdown occurs. PM means incorporation of a maintenance culture into the mainstream culture of the facility.

Every staff member has a responsibility to observe, monitor and report on equipment performance. After all they are the users and therefore the "field experts" and it means that a program is laid down where assets are periodically examined for potential failure.

It is not cyclic maintenance where you overhaul, ready or not, it is examination of performance, assessment of safety and discussion with users and technicians about whether the need for improvement exists.

Benefits of a PM program

The correct PM program gives you:

- Increased usage
- Better appearance
- Less failure
- Increased safety
- Fewer accident or insurance claims
- Increased productivity
- Reduced callout or ad hoc repair cost
- Parts when you need them.



Setting up a PM program

A PM program can be set up by:

- Calendar days, weeks or months - timing of a period for examination of the asset that is neither too often or infrequent
- Usage - the number of uses or cycles an asset has gone through. For example the common office scenario of the photocopier and the number of pages it has copied
- A combination of the two.



PM applies to any computer system asset.

PM leads to and overlaps with predictive maintenance, where you use history or criticality of an asset to predict ahead what it might do and you intervene by examination.



Activity 23 - Establish a PM schedule



You are asked to prepare a PM schedule for computer system assets.

COMPUTER SYSTEM ASSET	PREVENTATIVE MAINTENANCE ACTIVITIES REQUIRED
Daily	
Monthly	
Every 3 months	
Every 6 months	
Every year	

Scheduling of maintenance

Scheduling is the allocation of labour power, materials and equipment, at specific times and locations, for accomplishment of prescribed maintenance work.

Scheduling of asset maintenance can be undertaken in two phases, the first being an external assessment of the asset's operability and the second being the asset taken offline for preventive or routine maintenance.

The former should usually be undertaken with the users present as their perspective is invaluable, with staff able to comment on efficiency and inefficiency.

The second has to be timed more carefully as inevitably your overall facility operability will be reduced temporarily whilst the equipment is offline.

If your preventive program is well planned however, most scheduled maintenance should run complementary to operations.

A good preventive program means that in any long-term period of examination, such as one year, all computer assets have been inspected and assessed at the appropriate intervals and at the correct number of times.

Consider scheduling as synchronisation of:

- Labour available - in-house or contract
- Materials and parts available
- Special operation needs
- Contingencies allowed for sudden emergency call-backs
- Minimised impact on overall operability of the facility
- Most efficient cost
- Minimised customer awareness
- Minimum downtime
- Quick start-up and return to operability.



Maintenance request forms

Maintenance requests are used to report that something is faulty and has to be corrected.

They are a request for work to be done, usually maintenance or repair. The order must be given a priority rating.

Information required on a Maintenance request form include, but are not limited to:

- Date
- Description of problem
- Department responsible
- Name of person reporting problem
- Urgency.



Repairs

The repair process returns a faulty computer system asset into an operational condition.

Something is broken on the asset and the repair process makes the machine functional again.

Repairs could be considered as simple as replacing small parts, to the complete replacement of the asset because the required repairs are not economically feasible.

In most cases repairs imply some immediacy where action needs to be taken promptly to ensure the asset is in working order.

Unlike maintenance repairs are not predictable and cannot be planned.

There may also be an element of time pressure as the asset is not available and it may be needed.

This elevates the urgency of the repair and also increases the importance of reporting the issue to ensure that prompt action is taken.

In other cases a repair may not be immediate and the tools are still serviceable, although not as efficient before the damage. A monitor may have a cracked screen that makes the image hard to read but is still usable.

Maintenance and repairs without a time pressure component can be arranged to suit the needs of the business and match with the time available for service repair times. Users may be willing to suffer some inconvenience in return for use of the asset.

If a computer asset does require repair there will need to be a record maintained of the relevant service authority.

This can be a list of authorised repairers can be kept including contact names, numbers and people can be recorded.

In the next section we will explore in more detail the considerations involved in using external experts and technicians.



Physical cleaning of computer assets

An important part of ensuring computer system assets are maintained, hygienic and retain a professional and functional look, is through regular cleaning.

For most computer assets there will be internal cleaning procedures which are simple and straightforward, but have to be regularly performed.

Cleaning keyboard

- Unplug the keyboard from the PC
- Wipe keyboard with a damp cloth
- Many keyboards have keys that will come off if pulled - if this is the case, remove the keys, clean and replace them
- Standard operating procedure may be to cover the keyboard when not in use, with a plastic cover.

Cleaning monitors

- To clean the screen, first turn the PC off
- Use a soft cloth and clean water
- Dampen the cloth and clean the screen
- Do not spray chemicals or water directly onto the screen as it may penetrate behind the screen and do damage to the electronics.

Cleaning mouse

- Open mouse by undoing underside ring beneath the mouse ball
- Remove ball and clean
- Shake the mouse gently to allow dust and particles inside, to fall out
- Scrape built-up grime and dust from internal rollers
- Replace ball and ring.



Activity 24 - Identify physical cleaning procedures

Why is it important to regularly clean computer assets?

You are asked to identify the activities associated with physically cleaning computer assets.

COMPUTER SYSTEM ASSET	CLEANING ACTIVITY / TASK	CLEANING ITEMS REQUIRED

Cleaning of systems and internal memory

Besides conducting an actual clean, 'internal cleans' may include:

- Creating more space on the hard disk
- Reviewing programs
- Deleting unwanted files
- Cleaning dust from internal surfaces
- Backing up files before major maintenance
- Checking hard drive for errors
- Decrementing the hard disk
- Using up-to-date anti-virus programs.



Replacing consumables

In order for a tourism organisation to be operating efficiently, it is essential that not only can staff operate the computer system assets that exist within the office environment, but to also be able to identify and replace used technology consumables.

The most common type of maintenance required for new computer systems relates to the supply of consumables.

These are the items or materials that the assets consume during its normal operation.

Often these items involve an order process so a procedure needs to be established where a person is allowed to order the consumables and a source of supply has been established.

Although the detail for each tool will be different, the process involved is the same and once established for one tool can be followed for a range of tools.

When preparing systems for replacement of consumables the following needs to be considered:

- Where are the consumables stored?
- When should they be re-ordered?
- What should be ordered?
- What are common 'unit' sizes?
- How much should be ordered?
- Use of manufacturer products?
- Who should order?
- From whom should they be ordered from?
- Written record of consumables ordered?
- Ordering process and authorisation?



Training

There is also the issue of training. In some cases a local expert will install the replacement to ensure that it is installed and installed correctly.

As well as the physical process, there may be operational considerations, such as unscrewing screws, removing items, turning off equipment or performing back up of data, to name but a few tasks.

The process and procedure for replacement for each of the consumables for each of the assets needs to be documented and stored in an accessible location.

In some cases this documentation will be written in house to explain what and how to make the changes. In other cases it may be a hint list that is used in conjunction with the instructions on the tool.



Activity 25 - Identify consumables

You are asked to identify the different types of consumables that need to be replaced for business computer systems.



COMPUTER SYSTEM ASSET	REPLACEMENT ITEM	SUPPLIER	UNIT SIZE	COST

3.2 Report faults to the nominated person or supplier for rectification

1.3 Establish and implement procedures for identifying and resolving performance problems including use of specialised assistance

Introduction

Regardless of how much preventative maintenance you perform on your computer system assets, at times they will breakdown and faults will appear.

When faults appear, it is important to ensure that:

They are identified in a prompt manner

The correct procedures are in place to take the correct action to resolve the fault

All staff are expected to monitor the operation of equipment they work with.



When any of the computer assets shows signs there is a problem, this must be reported to the appropriate person as soon as possible. Timely reporting is important. The quicker the asset is checked out and fixed the better.

In this section we will look at the process of identifying faults and types of corrective action that can be taken to resolve faults.

Examples of faults

Common faults with computer equipment are usually listed in the Troubleshooting section of the relevant instruction manual.

Whilst the types of faults are endless examples of common faults include:

- Computer is too slow
- Internet downloads or uploads slow
- Computer won't start up
- Windows won't boot
- Computer keeps restarting on its own
- Monitor is blank
- Abnormally functioning operating system or software
- Strange noises
- Overheating
- Pop-up ads keep appearing



- READY light is not lit, or is glowing red
- Self-test labels will not print
- Labels will not feed, or feed continuously
- Flat batteries, or batteries have been removed
- Electricity cord has been accidentally knocked out or the plug has been removed
- Printer won't work
- Poor print quality
- Internet not connecting
- Wi-Fi keeps disconnecting
- Data or other connections have worked loose, or dropped out
- Can't open email attachments
- Can't email other people
- Programs aren't working on your PC
- Power point is switched 'on' but the machine is turned 'off'
- Scanner not reading at all, reading partially, or reading incorrectly
- Lack of signal from scanner to register/terminal – host stops responding
- Toner or ink has run out
- Paper-rolls have run out, or paper has jammed
- EFTPOS landline connection cannot be made to the bank, or connection is very slow
- Equipment malfunction or breakdown
- Power failure
- Operator error has caused the system or equipment to freeze
- Screen is frozen.



It is important to determine the standard operating procedure in each of the above situations to identify the action deemed appropriate.



Activity 26 - Identify common faults

You are asked to identify common computer system faults and research actions that can be taken by staff to remedy the problem before seeking external assistance.

You may use examples mentioned above.



TYPE OF COMMON FAULT	IN-HOUSE REMEDY TO RESOLVE FAULT

Identifying or anticipating faults

There are a number of ways to identify computer hardware or software faults or problems.

These methods are identified below.

The preparation of a scheduled maintenance program

The importance of having a PM program has been discussed in detail in Section 3.1 of this manual.

Routine checking of equipment

These can be visual inspections by office staff to determine whether or not there is a problem. In many cases, computer equipment will give notice of its imminent demise by making strange and unusual noises or functions not working as effectively as they should.

Reading the instruction booklets

These booklets should be filed away in some central place where all staff can have ready access to them. All instruction books or manuals will contain a 'troubleshooting guide' in them that can prove invaluable.

Encouraging feedback from colleagues

It seems to be fact that all equipment fails at the most inopportune time, at a time when we can least afford it to go down.

It is important that colleagues report any problems as soon as possible.

It should be standard operating practice to check with the people who are using the computers for their feedback. Very often, the problems can be 'people problems' and not really hardware or software issues.

Taking some time to talk to the staff and getting them to explain their problems can often indicate that it is not servicing or repairs that are called for. Instead staff training may be required.



Keeping a log book of detected faults

Each department should have its own register for staff to record problems that they have encountered with computer equipment and software.

This can act as a guide for problem analysis when either your internal IT person takes a look at things or when your external service provider arrives. By maintaining a record of fault types, this can be useful information to refer to when deciding on the future of a computer asset. Repeating faults may indicate that an asset is being used beyond its specifications and that a more robust model is required.

Fault repeating may also indicate that the repair process has fixed the symptom. Some companies do not maintain records of faults but only action taken and they may not investigate for a deeper problem.

Performing regular OSH inspections

This inspection should be undertaken both by staff who are familiar with how they use the equipment and know what is expected of it, and by an accredited expert in that area.

The results of such inspections should be made available to staff and management, and action taken, where required, to remedy problems.

Actions to take when computer faults are identified

Staff are partially responsible for identifying faults with computer equipment, and reporting these faults in accordance with enterprise procedures.

Identification and action to resolve computer equipment faults is difficult for untrained technicians. Usually we know there's a problem when the machine stops working, when it starts making a strange noise, or when we get an error message.

Standard procedure should be to:

- Note details of the fault - see below
- Notify the appropriate in-house IT expert
- Check manual for trouble shooting hints
- Utilise help desk, on-line help, etc.
- Call in IT support for a service call
- Record notification of a service call - this is useful to ensure staff that action has been taken and avoids a phone call from each user that wants to use the machine
- Equipment should be removed from service as soon as a fault has been identified – to limit frustration, for safety concerns, to minimise further damage
- Equipment should be labelled clearly and obviously 'Out of Service' – so that staff don't try to use it, and to assist in future repairs, etc.
- Equipment should be stored in the appropriate out of service area – where applicable, such as an office area or designated area in the store or lay-by area
- Appropriate report fault paperwork should be completed and submitted to the appropriate person or department – the appropriate person may be your IT support or manager
- Some faults may have to be repaired off-site – which may involve returning the item to the supplier or forwarding it to an accredited repairer.
- In extreme cases, a new item may have to be purchased. When major repairs are required, management may buy a new item instead of repairing the old one.
- Where repairs have been effected on an item, it is important for this item to be monitored in case the repairs prove to be ineffective and the establishment may be able to claim under some form of warranty or guarantee.



Note details of a fault

It is important to note the details of the fault as it can be helpful for technicians to arrive at a solution faster.

Complaints such as “machine not working” or “cannot copy” do not provide any helpful information. It is always the case that detail is helpful e.g. “machine jams only when duplexing on A4” directs the technician's attention to a particular area.

Documentation may be necessary that encourages detail or the expert collects the details.

Some manufactures have a standard form that is designed to ensure the correct and most complete data is collected to enable the problem to be solved as soon as possible.

This could include details such as:

- Person's name
- Contact details
- Email address
- Sequence of events
- Asset name
- Model
- Finance officer
- Problem description
- Service contract number
- Serial number of machine
- Urgency of the problem
- Access hours.



Using external specialist to conduct maintenance and servicing

The servicing may be undertaken by the in-house IT person, or by a contracted service provider.

Where an outside contract service provider is used, their attendance will be known in advance, and is usually planned for. This can involve scheduling down time for equipment, and arranging alternative equipment where necessary.

Where systems are networked, this is usually a relatively simple process, but nonetheless one that requires preparation

Types of specialist assistance

Specialised assistance may include:

- External Information Technology (IT) support
- Use of commercial ‘help’ desks
- Manufacturer.

Service details to be recorded

Outside contractors should also leave a written/ or printed report detailing the following:

- Work that they have done
- Time they have taken
- Parts that they have replaced
- Faults they have identified
- Warranties relating to repairs.

No IT service provider should be allowed to simply inspect or repair equipment without being required to be accountable by virtue of an account explaining what was done and explaining the charges that were made.

Where specific items of equipment are being serviced or inspected, it is best to engage service personnel who are licensed and accredited by the equipment manufacturer or supplier.

In some cases, this may be a requirement under the terms of the warranty with a failure to use proper service technicians can void the warranty or guarantee.



Activity 27 - Identify external specialist assistance

You are asked to identify contact details of three external specialists that may be contacted to help repair or resolve computer system problems.



TYPE OF COMMON FAULT	CONTACT DETAILS OF APPROPRIATE SPECIALIST

3.3 Establish and monitor procedures for ensuring security of data, including regular back-ups and virus checks

Introduction

In many cases, whilst the breakdown of hardware is a nuisance, the loss of data may be detrimental to an organisation.

Besides conducting a range of preventative maintenance and fault resolution actions, there are a number of other tasks that can be performed that will minimise the effect and loss of information due to computer system breakages or by theft of data.

In this section we will explore the activities associated with saving and protecting information stored in computer systems.

Establishing passwords

In order to protect the integrity of the system from unauthorised access and also to maintain the confidentiality of establishment information and customer details, it is usual for passwords to be used.

The use of passwords is vital for each and every computer system.

This not only relates to PC's but other equipment including photocopies and phones.

Some equipment have passwords that deny access to the asset or some function within the tool. The copier may have passwords for each department and another password that restricts access to the management of the copier, and the setup.

Some assets require passwords to be changed on a regular basis and users must be created to devise new passwords without duplicating an earlier version.



It is obviously important that both management and individual user passwords are recorded and secured.

The password is designed to control access so if the password becomes public knowledge it is useless as a password and quickly becomes a passport.

Password and user ID sequences and protocols

Before any task can be performed on any computer system asset, the operator must access the system.

Each person has their own dedicated password and may also have a user identification.

Access can be restricted via this password to certain areas of the system.

For example, owners or management will have full access to all areas, as Administrator for the system, whereas receptionists may be locked out of the 'Reports' and 'System Configuration and Setup areas'.

Passwords commonly require the user to enter their name in a predetermined manner, which may be upper and lower case sensitive, plus their initials or ID, and then their unique 'ID' or 'User' number when they log on.

Some systems will require one or more of these steps to be repeated as a means of confirming the authenticity of the operator before log in is granted.

Successful completion of this process allows the individual access to the system at the levels they have been allocated and tracks their usage of the system.

Operators should log out when they finish their shift or go for a break so as to prevent unauthorised access via their password to an 'open' system.

Security and privacy SOP's

Establishments may have Standard Operating Procedures (SOPs) in relation to their CRS that require passwords to be changed on a regular basis in order to help maintain security.

For example, you may be required to change your password every 30 days. It should be standard practice:

- Not to reveal your password to anyone
- Not to write your password down anywhere near your terminal
- Not to allow anyone to access the system using your password
- To immediately change your password if you believe it has been compromised and notify management accordingly.

Keys to creating passwords

- Passwords must consist of at least seven characters and must be a mixture of alphabetic or numeric characters
- Passwords must be chosen by the user and kept in confidence
- Only seven attempts to log-in to the ABC Network will be allowed
- Passwords must be changed immediately if discovery is detected or suspected
- Passwords cannot be reused in ABC Network and applications
- Passwords must not be predictable from knowledge of the user or from the previous password
- The system will automatically require users to change their passwords every 45 days.



Prevent computer viruses

What is a computer virus?

Computer viruses are identified as small software programs that have been designed to spread from one computer to another and to negatively interfere with the operations of a computer.

There are endless computer viruses that exist, each with a detrimental purpose including:

- Corrupt or delete data on your computer
- Use your email program to spread itself to other computers
- Erase all information on hard disks.

The spread of computer viruses

One way to control viruses is to identify how viruses can be spread.

The way computer viruses can spread includes:

- Through internet downloads
- As attachments in email messages or instant messaging messages
- Disguised as attachments of audio and video files and images

The main aim of controlling viruses is to:

- Remove it
- Prevent further infection.

Types of virus preventative actions

Examples of virus preventative actions include, but certainly not limited to:

- Install antivirus tools
- Keep computer current with latest updates
- Run regular virus checks
- Run regular scandisk facility.





Activity 28 - Identify antivirus and security software

Conduct research and identify two possible antivirus and security software options to protect your business computer system.

OPTION 1

Name of software	
Advantages / key features of software	
Disadvantages of software	
Number of PC's / equipment protected by software	
Cost of software	
Method to download software	

OPTION 2

Name of software	
Advantages / key features of software	
Disadvantages of software	
Number of PC's / equipment protected by software	
Cost of software	
Method to download software	

Use computer systems in a safe and secure manner

It is important that staff use business computer assets not only according to organisational requirements, but also in such a way so as to promote a safe and secure work environment.

The exact nature of the organisational requirements will vary between organisations, and alter with the degree and nature of the work to be done.

Several general organisational requirements include:

- Use the technology in the correct manner – this means staff are required to use the equipment or software as it was designed to be used. Training or an in-depth reading of the relevant manuals, coupled with an appropriate amount of practice is essential
- Use the technology effectively – you are required to utilise the technology to the best of your expertise and capabilities
- Maintain security – through passwords, secure storage of important information and hard drives
- Staff should not use equipment for private use - this extends to surfing the internet, sending private emails, performing private work during work hours or on workplace equipment
- Do not use private discs or flash drives in workplace computers – to help reduce the chance of virus infection of the workplace system
- Never allow customers or others to access or use establishment equipment, unless the equipment is specifically for customer use
- Report any instances of breaches of protocols immediately to management.



Perform regular backups of data

Despite our best efforts, despite all our preventative maintenance, and despite all the feedback in the world from staff, technology will still let us down.

One good way of minimising the adverse impacts of the computer crashing is to do regular backups of data.

A back-up is a second copy made to protect your information against accidental loss or damage. Many establishments back-up their records and files on a daily basis.

Doing back-up means you can restore lost information quickly and with confidence so that you return to the point where you were before the loss occurred.

The loss of information may be caused by human error, sabotage, technical failure, theft or even a major incident such as a fire in the premises.

Back-up copies of all important programs and data should always be maintained. It must become standard operating procedure to back-up files, including the hard drive.

The higher the value of the data, the more important it is to the organisation, therefore the more comprehensive the back-up system should be.

Golden rules to backing up data

A simple way to minimise the loss of information is to have three versions:

- Original
- Back up of the original
- Back up of the back-up.

In addition it is vital to have back up versions stored in different secure locations.



Methods to back up data

Following are common methods used to back up valuable computer data:

- Online storage services - such as iDrive, iCloud, Dropbox, Carbonite, Google Drive, Mozy or SkyDrive
- Consolidate data on different PC's and phones to one centralised point
- External hard drives
- Using backup software - such as Windows Backup.



Activity 29 - Identify online storage services

Conduct research and identify a suitable online storage service to store your business computer system data.

Identify the advantages of using an online storage service.



Provide a summary of the online storage service you feel is most appropriate for your organisation:

Name of online storage service	
Advantages / key features	
Disadvantages	
Costs	

3.4 Create document filing systems and procedures to reflect the size, nature and complexity of the workplace

Introduction

In the previous section we explored the different methods to ensure the security of computer data.

In this section we will identify the different policies and procedures that may be implemented in managing and storing data.

Types of data management policies and procedures

All data entered into any computer equipment must be entered in accordance with organisational policies and procedures.

These policies and procedures may relate to:

- Administration requirements
- Internal accounting systems
- Authorised use of equipment
- Security
- Confidentiality
- Timing and frequency requirements.



Storing data

It takes lots of time, money and effort to enter data into computers, and that information must be safeguarded.

If the data is lost it is highly unlikely that it can all be remembered, or that it will all be able to be reproduced.

With this in mind, it is easy to understand that appropriate storage is critical.

While many operators are competent in opening, saving and closing files, it is alarming how few people are familiar with enterprise requirements for storing data.

They seem to believe that it is only other people's computers that crash, only other people's computers that get attacked by viruses, become corrupted or get stolen or even 'perish' in a fire.

The workplace reality is that storing data is as important as generating it in the first place.

After creating or amending files, documents and records it is important that you are able to store data appropriately, and then exit the applications without damage, and without loss of data.

This storage of data may include:

Storage in directories and sub-directories

These may be allocated random names but most large establishments have specified policies and procedures regarding this.

Storage on hard drives

This is the most common form of saving and storing data on the computer's hard drive. Most premises do this, and many small operations don't bother backing-up this option. If there is a fire, theft, industrial espionage or a simple hard drive crash or corruption the results can be catastrophic.

Storage on networked computers

Some properties back-up data on other hard drives within the network. This means individual PC's or terminals will send their data to a centralised point, via a networked application, for storage there.



Appropriate filing and storage of hard copies of computer-generated documents

In some cases a hard copy of data may be taken and filed or archived for future use.

While hard copies cannot be readily amended they at least provide a source document for future reference and use.

Hard copies may not be very 'high tech' but it is amazing how valuable they become when the hard drive crashes, or there is a power failure and the data from the hard drive is unavailable.

Establishing data management

Data Management is the professional discipline and organisational function of managing and maintaining full and accurate data records to meet operational business needs and accountability requirements.

These guidelines aim to ensure correspondence is appropriately listed in the electronic filing system according to the organisational classification system. The official title given to all files should be classified using the organisational classification system and should enable a summary or concise statement of the contents of the file to be made.

Using the classification system and limiting the use of acronyms, abbreviations and punctuation will subsequently aid in the successful retrieval of all files when requested

These are policies to ensure that XYZ creates and maintains full and accurate records as evidence of its business activities and transactions.

Key outcomes of a data management system

- All data will be managed in accordance with set organisational procedures
- All staff understand what constitutes official data
- Business activities are to be documented in the form of records
- Any data or record can be retrieved easily when requested
- All data must be stored safely and securely

- Archived data will be managed in accordance with legislative requirements
- Intellectual Property is captured, maintained and protected.



Activity 30 - Identify importance of data management systems

Explain the importance of having a data management system in place.

What are examples of policies and procedures you would have in place to ensure staff store data correctly?

Policies	
Procedures	

Work Projects

It is a requirement of this Unit you complete Work Projects as advised by your Trainer. You must submit documentation, suitable evidence or other relevant proof of completion of the project to your Trainer by the agreed date.

3.1 Please complete the following activities relating to this Performance Criteria:

- Activities 23,24,25

3.2 Please complete the following activities relating to this Performance Criteria:

- Activities 26,27

3.3 Please complete the following activities relating to this Performance Criteria:

- Activities 28,29

3.4 Please complete the following activities relating to this Performance Criteria:

- Activity 30

3.5 Please complete the following activities relating to this Performance Criteria:

- Activities 19,20
-

Summary

Maintain computer equipment and systems

Establish and monitor systems for cleaning, minor maintenance and replacement of consumables in accordance with manufacturer's instructions

- Role of management in maintaining computer system assets
- Importance of conducting routine maintenance
- Preventive maintenance
- Scheduling of maintenance
- Repairs
- Physical cleaning of computer assets
- Cleaning of systems and internal memory
- Replacing consumables

Establish and implement procedures for identifying and resolving performance problems including use of specialised assistance

Report faults to the nominated person or supplier for rectification

- Examples of faults
- Identifying or anticipating faults
- Actions to take when computer faults are identified
- Note details of a fault
- Using external specialist to conduct maintenance and servicing

Establish and monitor procedures for ensuring security of data, including regular back-ups and virus checks

- Establishing passwords
- Prevent computer viruses
- Use computer systems in a safe and secure manner
- Perform regular backups of data

Create document filing systems and procedures to reflect the size, nature and complexity of the workplace

- Types of data management policies and procedures
- Storing data
- Establishing data management

Presentation of written work

1. Introduction

It is important for students to present carefully prepared written work. Written presentation in industry must be professional in appearance and accurate in content. If students develop good writing skills whilst studying, they are able to easily transfer those skills to the workplace.

2. Style



Students should write in a style that is simple and concise. Short sentences and paragraphs are easier to read and understand. It helps to write a plan and at least one draft of the written work so that the final product will be well organised. The points presented will then follow a logical sequence and be relevant. Students should frequently refer to the question asked, to keep 'on track'. Teachers recognise and are critical of work that does not answer the question, or is 'padded' with irrelevant material. In summary, remember to:

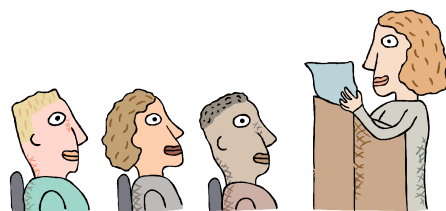
- Plan ahead
- Be clear and concise
- Answer the question
- Proofread the final draft.

3. Presenting Written Work

Types of written work

Students may be asked to write:

- Short and long reports
- Essays
- Records of interviews
- Questionnaires
- Business letters
- Resumes.



Format

All written work should be presented on A4 paper, single-sided with a left-hand margin. If work is word-processed, one-and-a-half or double spacing should be used. Handwritten work must be legible and should also be well spaced to allow for ease of reading. New paragraphs should not be indented but should be separated by a space. Pages must be numbered. If headings are also to be numbered, students should use a logical and sequential system of numbering.

Cover Sheet

All written work should be submitted with a cover sheet stapled to the front that contains:

- The student's name and student number
- The name of the class/unit
- The due date of the work
- The title of the work
- The teacher's name
- A signed declaration that the work does not involve plagiarism.

Keeping a Copy

Students must keep a copy of the written work in case it is lost. This rarely happens but it can be disastrous if a copy has not been kept.

Inclusive language

This means language that includes every section of the population. For instance, if a student were to write 'A nurse is responsible for the patients in her care at all times' it would be implying that all nurses are female and would be excluding male nurses.

Examples of appropriate language are shown on the right:

Mankind	<i>Humankind</i>
Barman/maid	<i>Bar attendant</i>
Host/hostess	<i>Host</i>
Waiter/waitress	<i>Waiter or waiting staff</i>

Recommended reading

Chambers, Mark; 2013 (6th edition); *PCs All-in-One for Dummies*; For Dummies

Clements, Alan; 2006 (4th edition); *Principles of Computer Hardware*; Oxford University Press

Englander, Irv; 2014 (5th edition); *The Architecture of Computer Hardware, Systems Software, and Networking: An Information Technology Approach*; Wiley

Gookin, Dan; 2011 (2nd edition); *Troubleshooting and Maintaining Your PC All-in-One for Dummies*; For Dummies

Kurose, James; 2012 (6th edition); *Computer Networking: A Top-Down Approach*; Pearson

Levine, John; 2011 (11th edition); *The Internet for Dummies*; For Dummies

Lowe, Doug; 2009 (9th edition); *Networking for Dummies*; For Dummies

Miller, Michael; 2014 (7th edition); *Computer Basics Absolute Beginner's Guide*; Que Publishing

Patt, Yale; 2003 (2nd edition); *Introduction to Computing Systems*; McGraw-Hill Education

Stokes, Abby; 2012 (Revised edition); *Is This Thing On?, revised edition: A Computer Handbook for Late Bloomers, Technophobes, and the Kicking & Screaming*; Workman Publishing Company; Revised edition

Trainee evaluation sheet

Monitor and maintain a business computer system

The following statements are about the competency you have just completed.

Please tick the appropriate box	Agree	Don't Know	Do Not Agree	Does Not Apply
There was too much in this competency to cover without rushing.	☐	☐	☐	☐
Most of the competency seemed relevant to me.	☐	☐	☐	☐
The competency was at the right level for me.	☐	☐	☐	☐
I got enough help from my trainer.	☐	☐	☐	☐
The amount of activities was sufficient.	☐	☐	☐	☐
The competency allowed me to use my own initiative.	☐	☐	☐	☐
My training was well-organised.	☐	☐	☐	☐
My trainer had time to answer my questions.	☐	☐	☐	☐
I understood how I was going to be assessed.	☐	☐	☐	☐
I was given enough time to practice.	☐	☐	☐	☐
My trainer feedback was useful.	☐	☐	☐	☐
Enough equipment was available and it worked well.	☐	☐	☐	☐
The activities were too hard for me.	☐	☐	☐	☐

The best things about this unit were:

The worst things about this unit were:

The things you should change in this unit are:

Trainee self-assessment checklist

As an indicator to your Trainer/Assessor of your readiness for assessment in this unit please complete the following and hand to your Trainer/Assessor.

Monitor and maintain a business computer system

		Yes	No*
Element 1: Monitor effectiveness of computer equipment			
1.1	Monitor and evaluate the effectiveness of computer equipment to ensure it meets enterprise needs through ongoing consultation with users	☐	☐
1.2	Monitor effectiveness through assessment of productivity and efficiency	☐	☐
1.3	Establish and implement procedures for identifying and resolving performance problems including use of specialised assistance	☐	☐
1.4	Identify computer training needs and provide training opportunities	☐	☐
Element 2: Purchase hardware and software			
2.1	Identify broad computer equipment hardware and software requirements accurately and research possible new approaches using appropriate information sources	☐	☐
2.2	Assess the feasibility of acquiring or upgrading hardware and software based on available information	☐	☐
2.3	Evaluate options against the needs of the enterprise, including advantages and limitations	☐	☐
2.4	Investigate user-friendliness, system support and training issues	☐	☐
2.5	Seek external specialist assistance as required	☐	☐
2.6	Make decisions on computer acquisitions in consultation with appropriate colleagues	☐	☐
2.7	Plan introduction of new computer systems to take account of impacts on colleagues and customers	☐	☐
2.8	Ensure staff are adequately trained in using new systems	☐	☐
2.9	Monitor new systems for efficiency and take action accordingly	☐	☐

		Yes	No*
Element 3: Maintain computer equipment and systems			
3.1	Establish and monitor systems for cleaning, minor maintenance and replacement of consumables in accordance with manufacturer's instructions	☐	☐
3.2	Report faults to the nominated person or supplier for rectification	☐	☐
3.3	Establish and monitor procedures for ensuring security of data, including regular back-ups and virus checks	☐	☐
3.4	Create document filing systems and procedures to reflect the size, nature and complexity of the workplace	☐	☐
3.5	Set up equipment to accept the software and functions being used	☐	☐

Statement by Trainee:

I believe I am ready to be assessed on the following as indicated above:

Signed: _____

Date: _____

Note:

For all boxes where a **No*** is ticked, please provide details of the extra steps or work you need to do to become ready for assessment.

William
Angliss
Institute

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